SND Mobile Design Bootcamp

Jeremy Gilbert & **Dave Stanton**

The Freedom Forum,
Nashville, TN 2011

About Us

The Plan

Today	
9AM	Registration
9:30AM	Introductions
10AM	UI Gestures
11AM	Information Architecture
12:30PM	Lunch
1:30PM	Proprietary vs Web Apps
2:30PM	Mobile Frameworks


GUITAR WORLD
5 SONGS
WITH BASS LINES
Children of Bodom
"LAKE BODOM"
Dethklok
"THUNDERHORSE"
Avenged Sevenfold
"ALMOST EASY"
Foghat
"SLOW RIDE"
Heart
"BARRACUDA"
guitarworld.com
Perry & Slash
SHOW YOU HOW TO BECOME A REAL
GUITAR HERO
WITH SPECIAL GUEST HEROES George Lynch • Joe Satriani • Alexi Laiho
PLUS: EXTREME HEROICS WITH JOHN PETRUCCI
PAUL GILBERT INTERVIEWS ROBIN TROWER
ALL THE ESSENTIAL LICKS, TRICKS & LOOKS!
Future
PLUS: A sneak preview of Guitar Hero: Aerosmith




Carrier
7:55 AM
100%
CNN
The New York
Photos
Safari
Settings
Contacts
Apple
LickOfTheDay
GUITAR WORLD
LICK OF THE DAY
Powered by TabToolkit
Developed by
Agile Partners

About You

- Your Name
- Your Smartphone (if you have one)
- Familiarity with HTML/CSS/Javascript:
    a. I read about HTML5 for fun
    b. I built my last site in tables
    c. What's this HTML thing?
- Where are you working/What do you do?

Mobile Design Bootcamp Goals

What do you want to learn?

What are we planning to cover

Smartphone Market

Smithsonian 1992:

- Phone companies are very excited about the prospect of something called the personal communication network, or PCN. This is a wireless pocket phone you could take to any city and be in constant touch.

Smithsonian 1992:

- Just think: someday people might receive personal phone numbers at birth and carry them the rest of their lives. Much of the basic technology needed to create a global network of personal phones already exists.

Smithsonian 1992:

- Many in the industry think Americans might buy more than 100 million PCN telephones, says Ashok Ranade, a Bellcore manager in radio research. But expect some holdouts.

"A lot of my friends would abhor the idea of having a phone in their pocket all the time, says **John Davis**, vice president of network development for Bell Laboratories.

US Mobile Market



82%

of Americans use
a mobile phone

Pew Internet



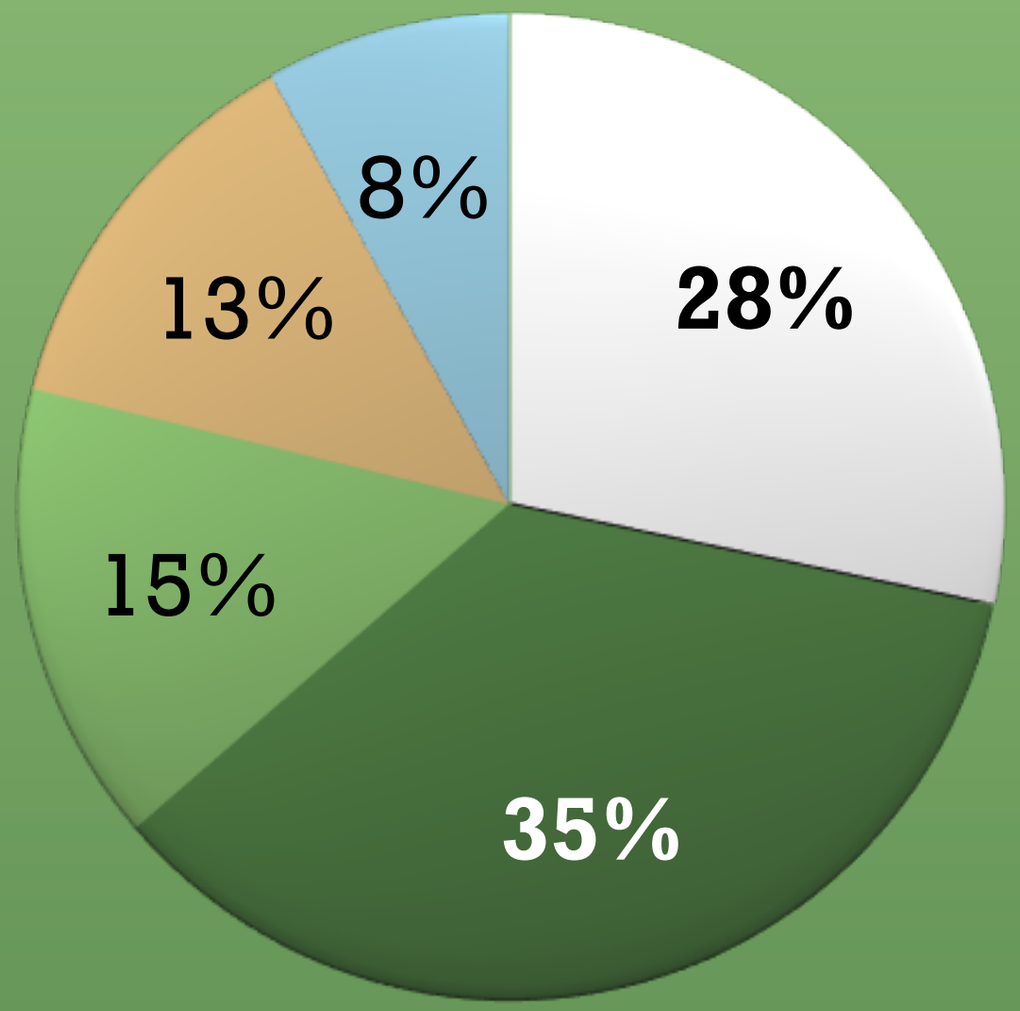
17%

of Americans use
smartphones

NielsenWire: Global Mobile

US Smartphones

- Apple
- Blackberry
- Windows
- Android
- Other



Smartphones Market Share By Operating System

[Mobile is Good]

Smartphone Apps

99.4%

of smartphone apps sold are by Apple



- Apple
- Others

Ars Technica

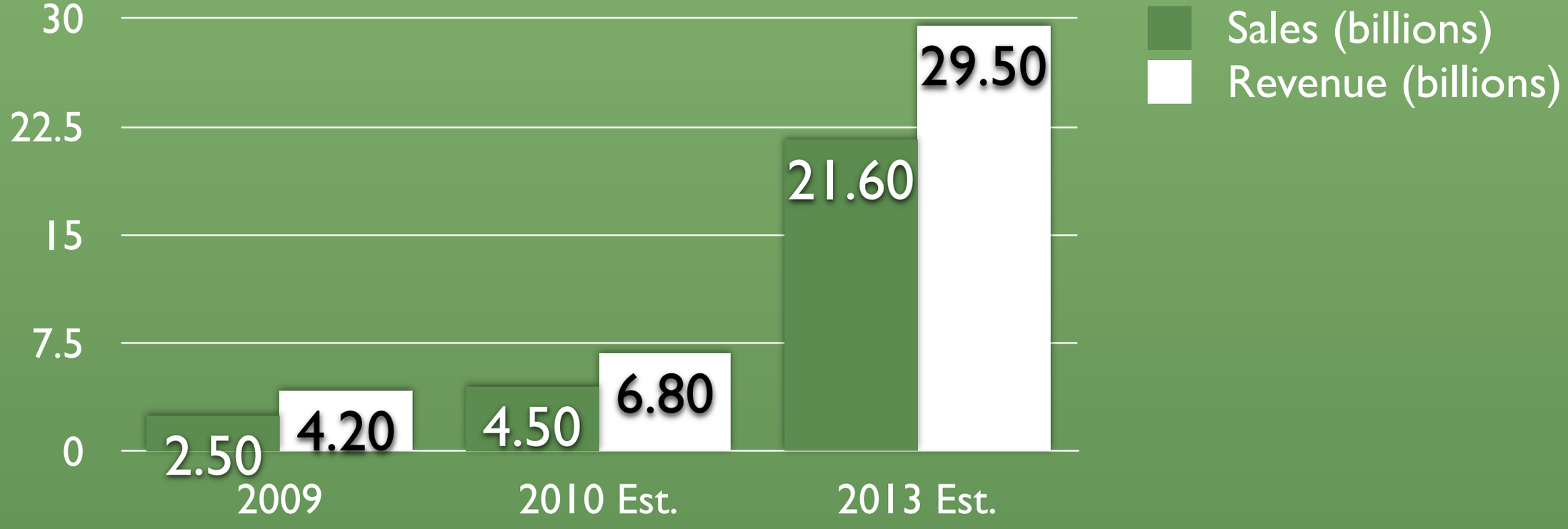
Apple Has Sold More Than

2 Billion

Mobile Applications

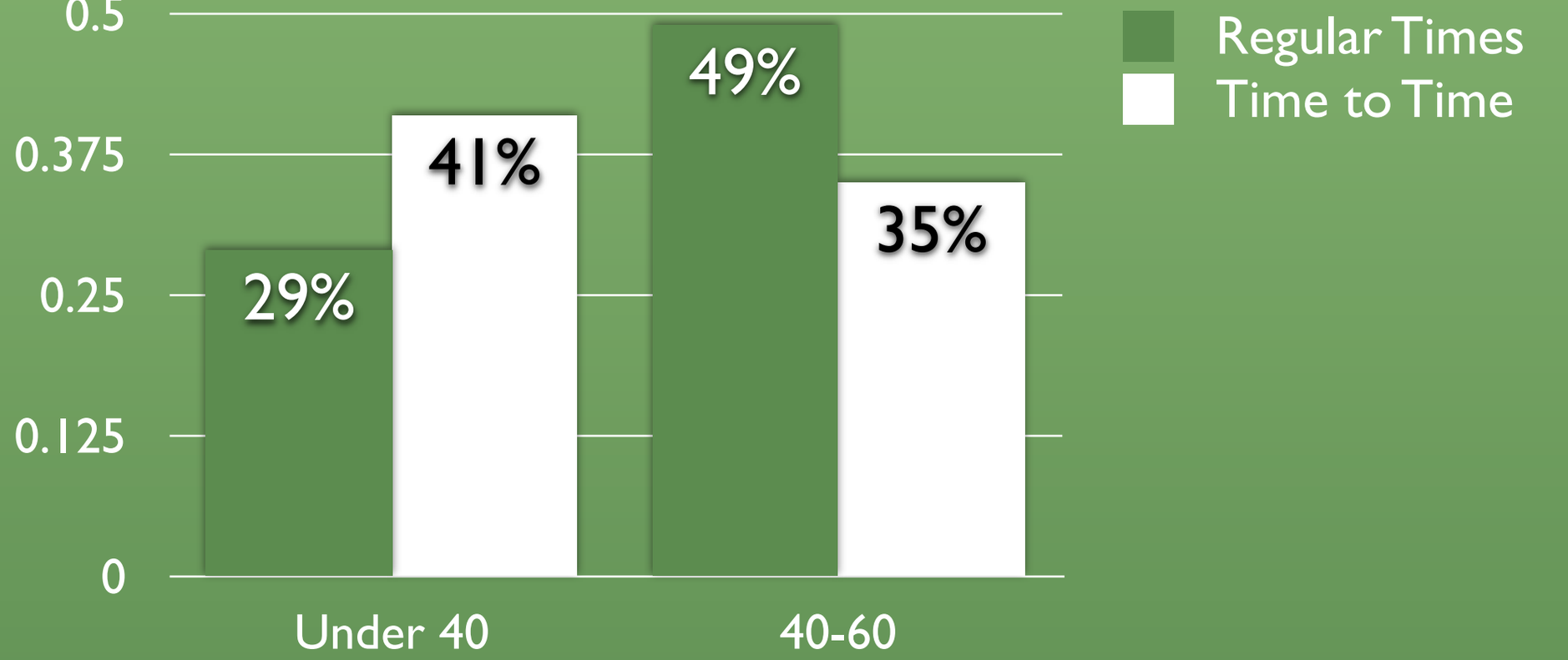
Ars Technica

Sales & Revenue



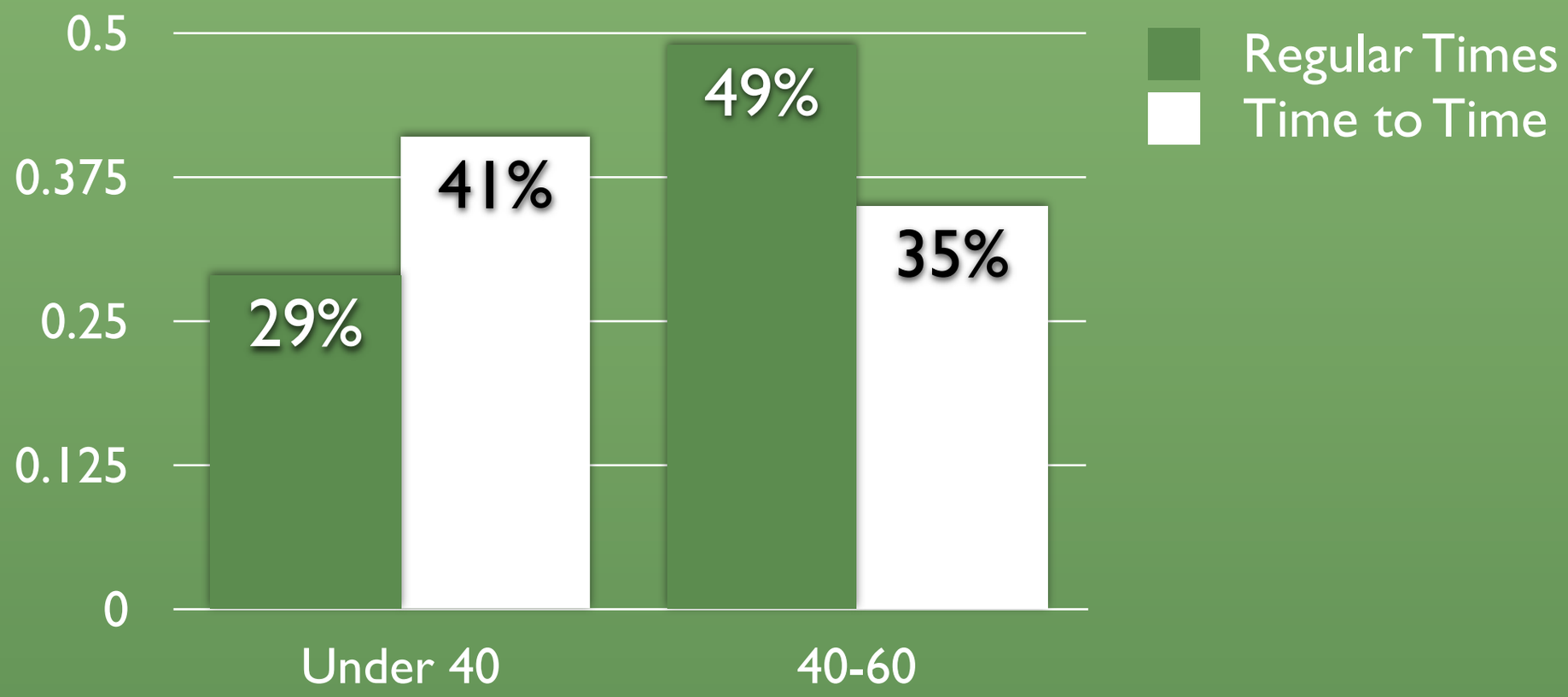
Ars Technica

News Readership Habits By Age



Pew Research Center: Online Papers Modestly Boost Newspaper Readership

Mobile News Habits



Pew Research Center: Key News Audiences Now Blend Online and Traditional Sources

When to Think Mobile

5. Hardware


Browse
abc NEWS
Consumer Reports Raises Doubts About Gold Firm...
Discounted Designer Shoes Too Good to Be True?
As Storm Brews, BP Relief Well Drilling Stopped
Baby Leopards Make Paris Zoo Debut
Toxic Makeup
Regulation
Cyberbullying: How to Talk to Your Kids
Shuffle
Back
yelp
Refresh
12. Nordstrom Cafe
9. California Crisp
7. Sarku Japan
1. Beach City Market
37 Reviews
6. Athens Market Downtown
0 Reviews
8. Great Khan's Mongolian...
11. O'Brothers
Restaurants
Bars
Everything

Remember:

1. Embrace device features
    a. Gyroscope
    b. Camera
    c. Voice Detection
2. Avoid gimmicks
3. Do not duplicate the web or the desktop

4. Geography


AT&T
10:30 PM
38 %
Chicago
Latest news
Filter
700-719 W. Chicago Ave.
Wednesday, July 21, 2010
Locations in the media
Chicago Real Estate Daily
Trading firm signs big expansion...
Trading firm signs big expansion at 600 W. Chicago
600 W. Chicago
Real estate listings
700 N Larrabee St #1113 listed on...
700 N Larrabee St #1113 listed on Trulia for $399,900, July 21
700 N Larrabee St
Crimes
Narcotics: Manufacturing/delivery...
Sidewalk. Reported 10:54 a.m. July 12.
AT&T
10:38 PM
92 %
Cities
Wrigleyville
Lincoln Park
Roscoe Village
Bucktown
Wicker Park
West Town
Chicago metropolitan area
Goose Island
Cabrini-Green
Near West Side
East Garfield Park
Dwight D. Eisenhower Expy
Grant Park
Chicago Harbor
Find cars by
Reserve
Drive
My Zipcar
AT&T
10:35 PM
93 %
foursquare
Jeremy Gilbert
MAYORSHIPS
0
BADGES
6
History
Friends
One River Place
Tue Jul 6
Hotel Monteleone
Fri Jul 2
Vaughan's Lounge
Thu Jul 1
Bacchanal Wine Bar
Thu Jul 1
Hotel Monteleone
Thu Jul 1
Friends
Places
Tips
Jeremy

Remember:

1. Take advantage of what you know about where users are
2. Think about what information users need on the go
3. Make it easy for users to access that information when they need it

3. Time


5:59 PM
Home
Fans Unfazed By New Armstrong Doping Alleg...
Floyd Landis told federal investigators doping was common on the U.S. Postal team.
+ Playlist
Appeals Court Strikes Down FCC Indecency Pol...
Under the policy, broadcasters face fines for even a single curse word on live TV.
Fine Art
Missing 'Priceless' Artwork? Call Robert Wittman
+ Playlist
Music News
Live Nation Stumbles Over Summer Concert Sales
Listen
Playlist
Hourly News
Programs
iPad
6:56 AM
npr
Home
news
arts & life
music
Done
Playlist
New
Do Dads Hate Parenting?
The recent New York Magazine article "All Joy, No Fun: Why Parents Hate Parenting," spar...
Blacks, Latinos Heavy Consumers Of Mobile Tec...
A recent Pew study indicates that African Americans and...
Racial Discussions In America: How Do We Ha...
The events surrounding the resignation of former U-S De...
The Tricky Ethics Of Video In A YouTube Era
Some media observers are questioning why important pla...
The White House Doctor
Dr. Connie Mariano has cared for some of the most prominent patients in the world. She wa...
History
What Digital Divide?
Clear Your Playlist
Export Boosts
But Cause Fear
French Firm's U.S
Angers Holocaust
Critics' Lists: Summer 2010
Fun In The Sun: Laugh-Out-Loud Summer Books
Favorite Sessions
Patty Larkin: Divine Aspirations
WFUV
+ Playlist

News Feed


Carrier
4:36 PM
NewsFeed
How hungry are you for news?
5 minutes
Snack
10 minutes
Fast Food
20 minutes
Full Meal
Back
Next


Carrier
4:36 PM
NewsFeed
The Full Meal is:
• 5 news briefs (appetizers)
• 4 full-length stories (entrees)
• 2 funny or light stories (dessert)
Coming First: Appetizers
Back
Next


Carrier
4:36 PM
NewsFeed
4/11
INTERNATIONAL
North Korea threatens nuclear and missile tests
SEOUL, South Korea - North Korea threatened Wednesday to conduct nuclear and missile tests and start an uranium-enrichment program in addition to its existing plutonium-based one, unless the U.N. apologizes for criticizing its recent rocket launch, dramatically raising its stake in the worsening standoff over its atomic programs.

Remember:

1. Fit the time users have available
2. Let users choose how much time to spend

2. Social


iPad
11:35 PM
Contents
Facebook
Recent contributors:
Ad Hudler
Laura Schulak Northrop
Chris Rukan
Josh En
Shared by Anne Glover – Peeps and friends, here's a great downtown St. Pete duplex to rent. Renee did a
iPad
11:39 PM
Contents
Twitter
Tampa Bay Rays aim to exorcise ghastly winless–in–Cleveland skid – St. Petersburg Times
Shared by pattycox
tampabay.com - Tampa Bay Rays aim to exorcise ghastly winless-in-Cleveland skid
By Marc Topkin, Times Staff Writer
In Print: Friday, July 23, 2010
Social Bookmarking
ADVERTISEMENT
Kids and digital storytelling: Who will teach them?
Shared by Storytelling News
ojr.org - By Robert Niles No comments | Archive Link
But it's not just my son. For many of his elementary school classmates, producing and distributing video has become as ubiquitous as writing and passing paper notes was to students of my generation. I've seen kids whip out Flip cameras and cellphones at school events and around the neighborhood, recording their
public school where more than half the students are receiving free or reduced-price lunches. Thanks to cheap cellphones and Flips, digital video technology has become so inexpensive that it's penetrated well beyond the middle class into poor segments of American society.
When I taught journalism in the university, I'd note with other instructors how the undergraduates

Remember:

1. It is much easier for users to discuss content than to create it
2. Find ways for users to interact with each other — centered on your content

1. Specificity



Remember:

1. Focus your app, have one primary use
2. Many small apps with loyal audiences are more valuable a single app with a fickle, large audience

Things to Avoid

- Unnecessary Fidelity
- Reinvent UI Cues
- Making Affordances Hard to Find/Understand
- Inconsistency
- Bad Error Messages or No User Feedback
- Asking Users to Tell You Things You Know



Designing for Small Screens

Gestures, Not Clicks

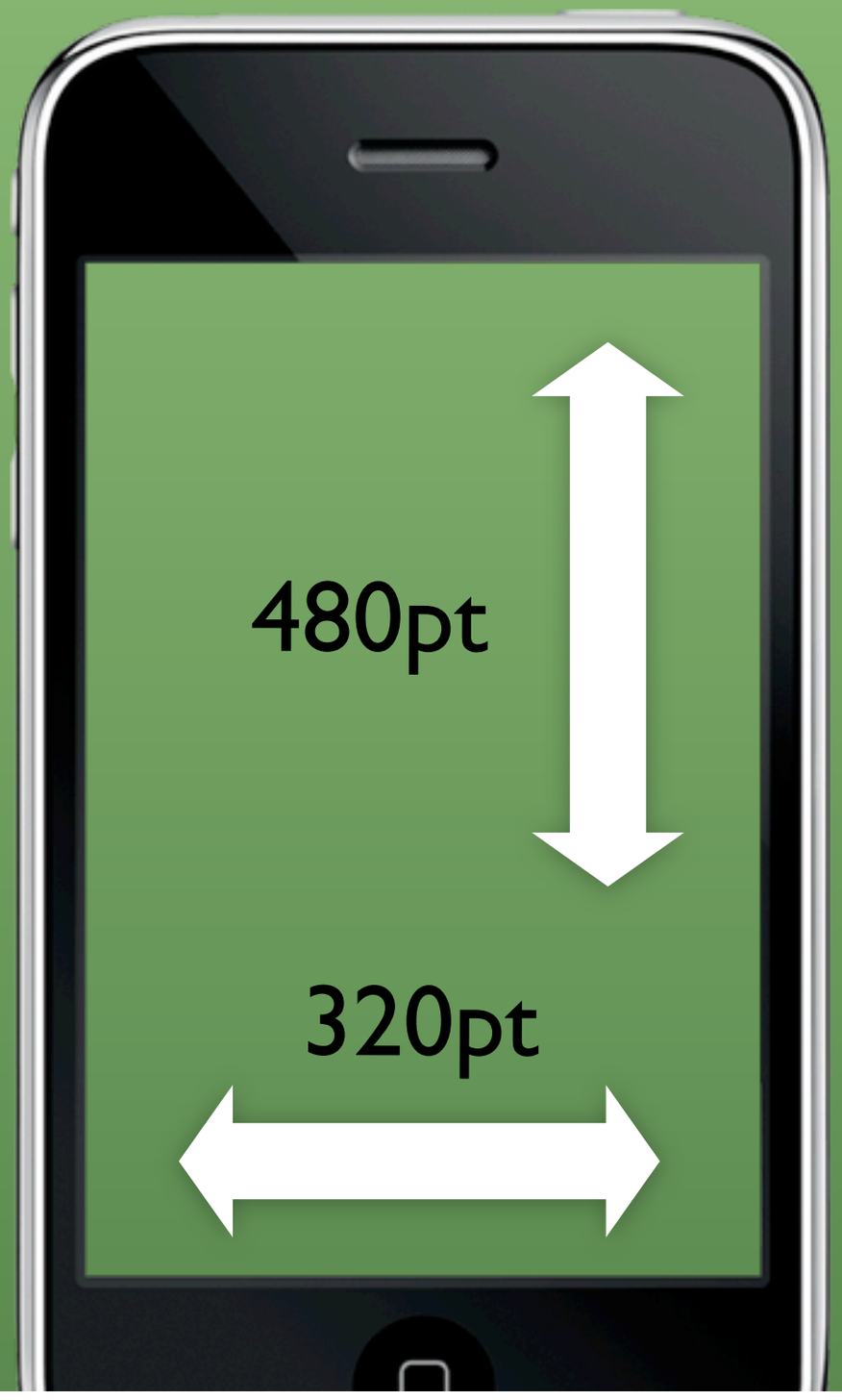
Gesture	Action
Tap	To press or select a control or item
Drag	Scroll or pan
Flick	Scroll or pan quickly
Swipe	In a table-view row, to reveal the Delete button
Double Tap	Zoom in and center or zoom out (if already zoomed in)
Pinch Open	Zoom in
Pinch Close	Zoom out
Touch & Hold	Edit text (select, select all, copy)
Shake	Undo

Apple Human Interface Guidelines

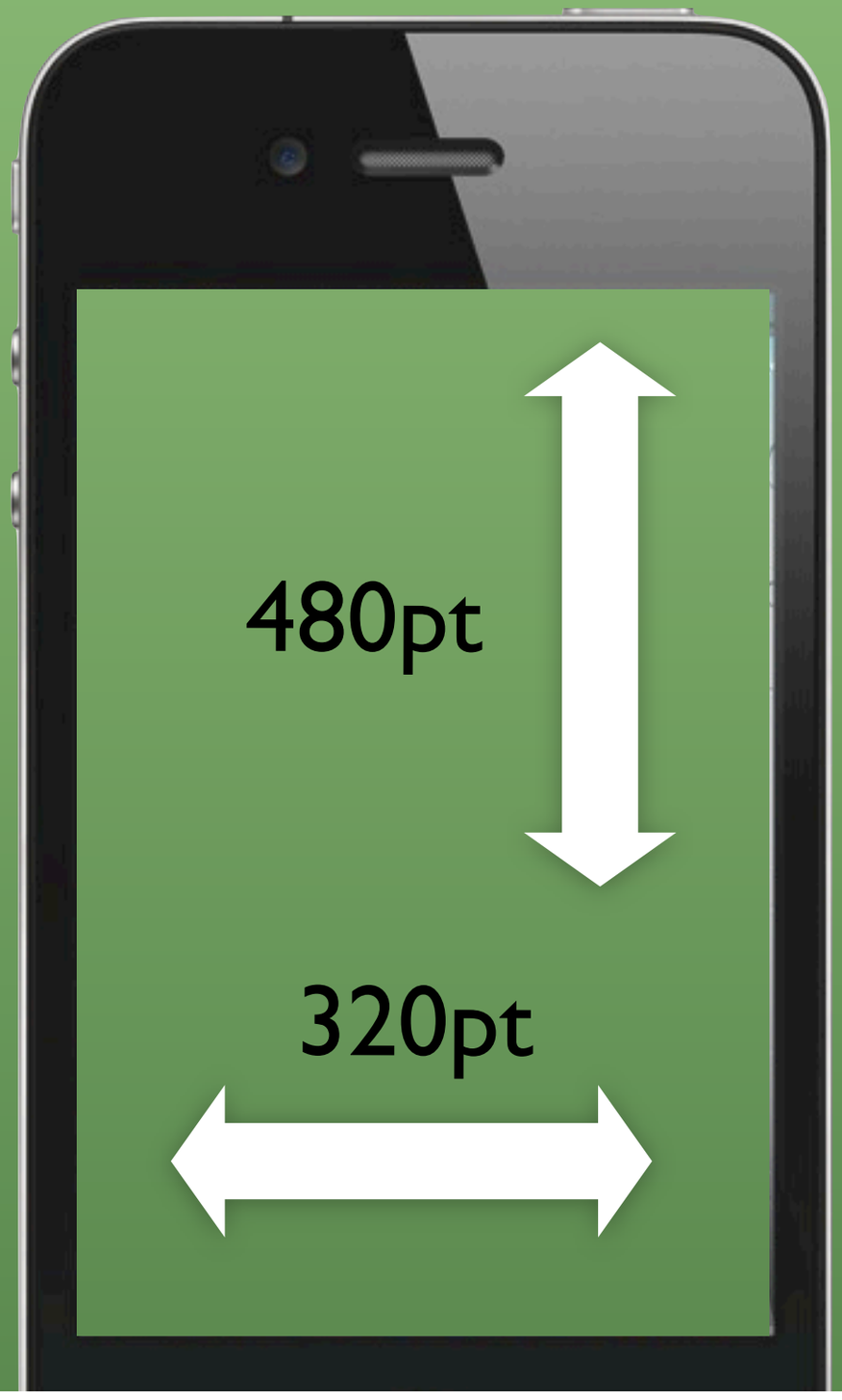
Points vs Pixels

- **Pixel** describes the size of a device screen or the size of an icon you create in an image-editing application.
- **Point** describes the size of an area that is drawn onscreen.

iPhone 3G



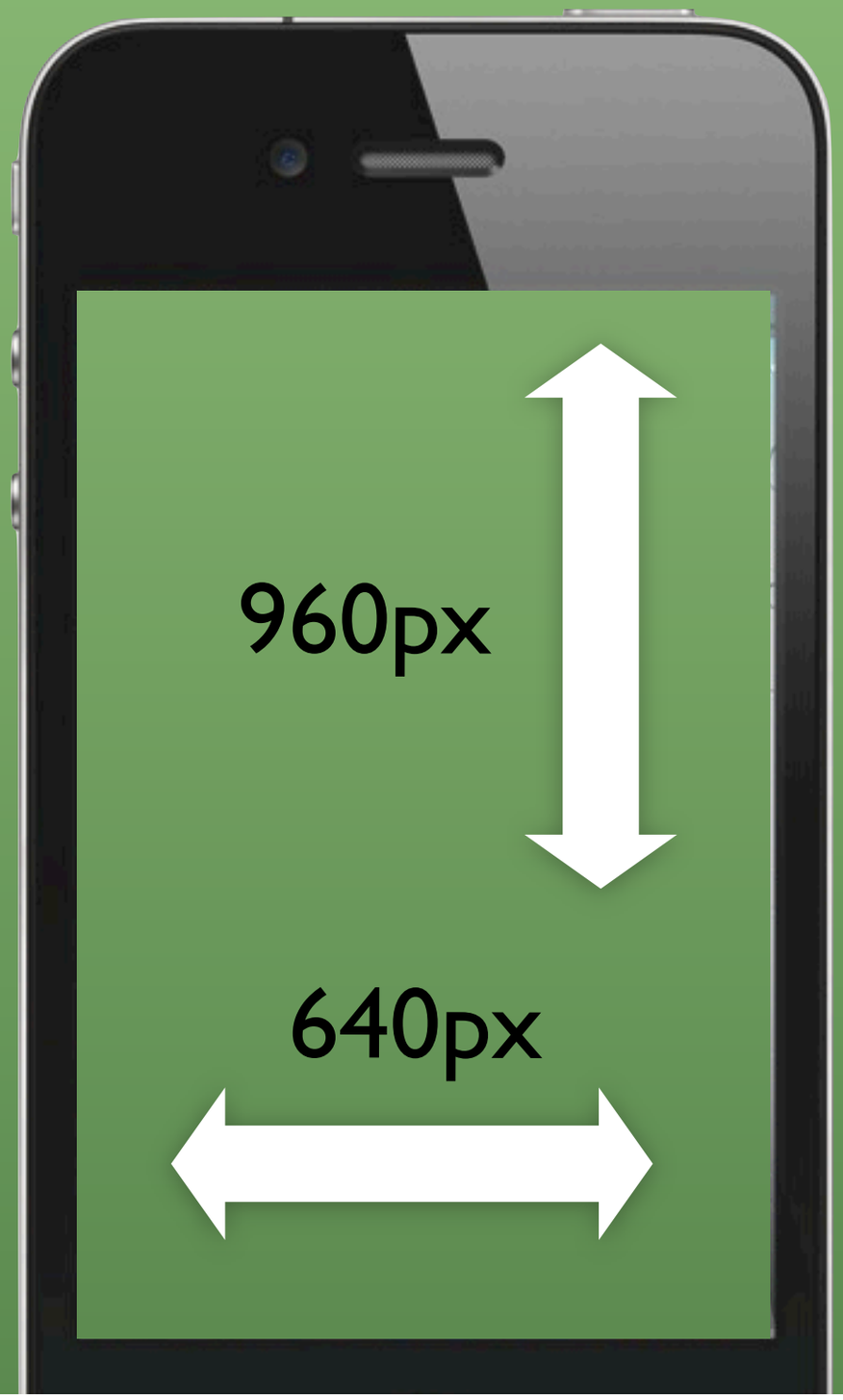
iPhone 4

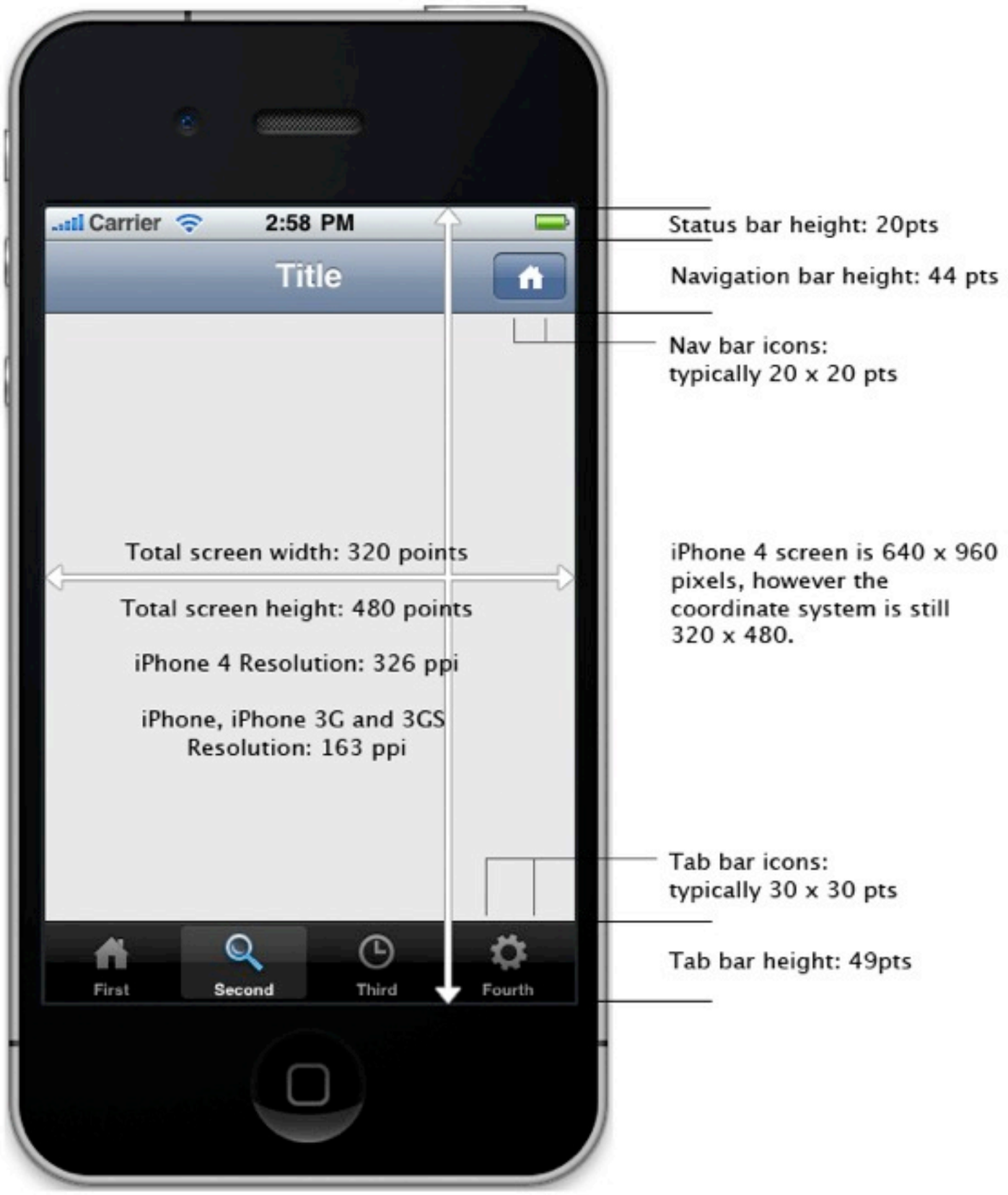


iPhone 3G



iPhone 4




Carrier
2:58 PM
Title
Total screen width: 320 points
Total screen height: 480 points
iPhone 4 Resolution: 326 ppi
iPhone, iPhone 3G and 3GS
Resolution: 163 ppi
First
Second
Third
Fourth
Status bar height: 20pts
Navigation bar height: 44 pts
Nav bar icons:
typically 20 x 20 pts
iPhone 4 screen is 640 x 960
pixels, however the
coordinate system is still
320 x 480.
Tab bar icons:
typically 30 x 30 pts
Tab bar height: 49pts

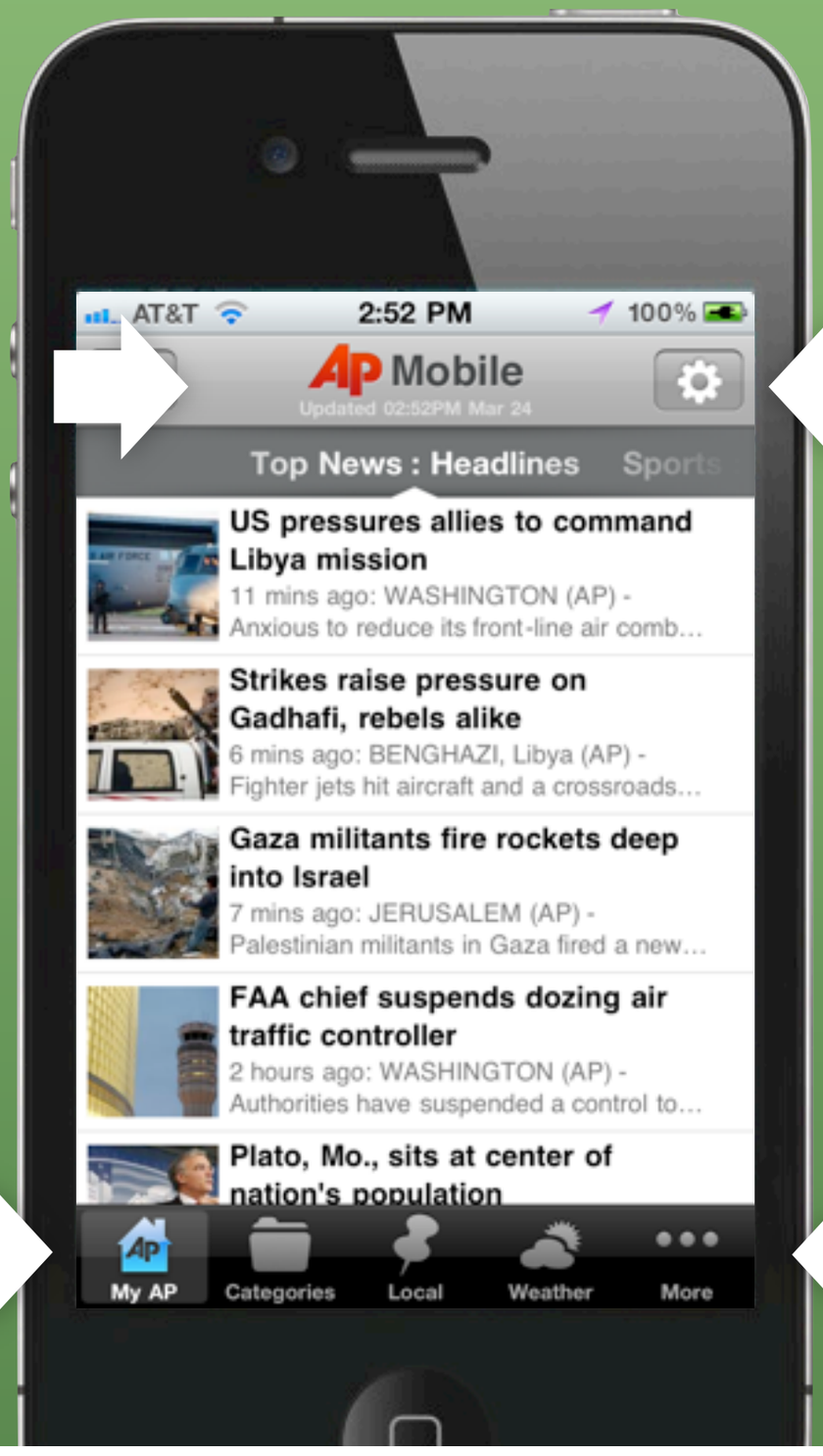
Finger-Sized

44pts



UI Conventions

Location



Settings

Primary Nav

Additional Nav

~~Multitasking~~

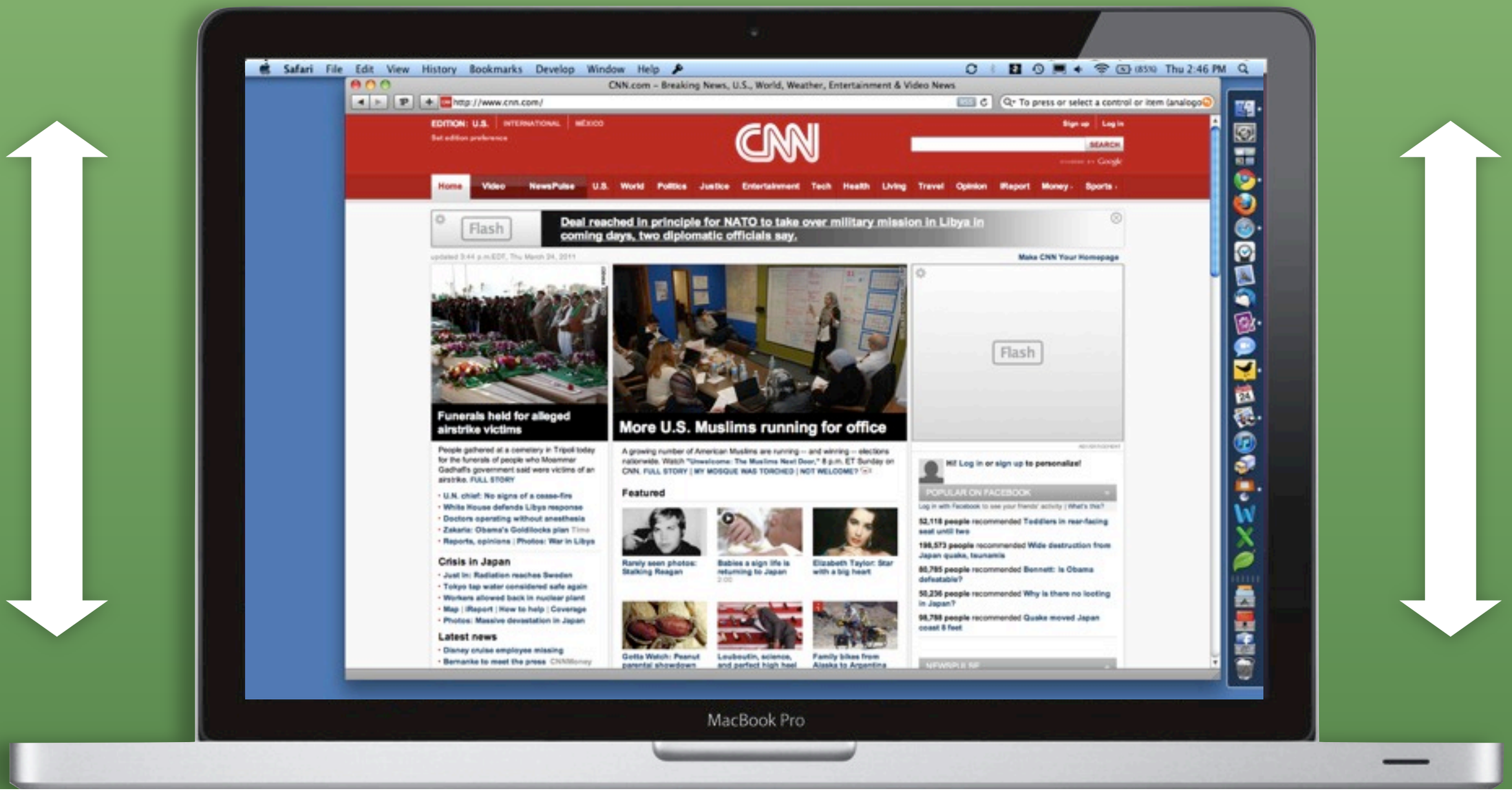


[Apple Human Interface Guidelines]



Cards vs Scroll

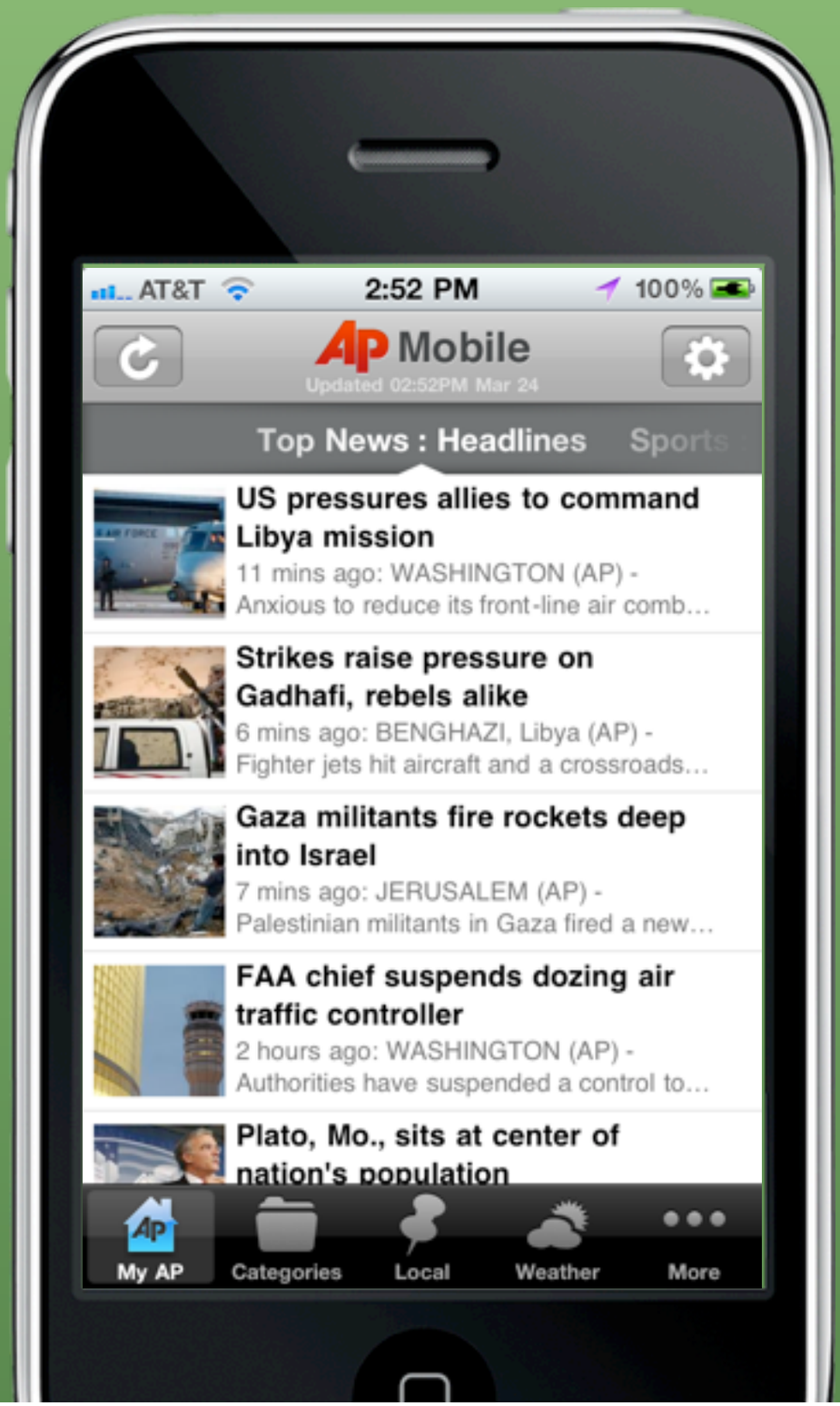
Nielsen Norman Group



Cards vs Scroll



Cards vs Scroll



Cards vs Scroll

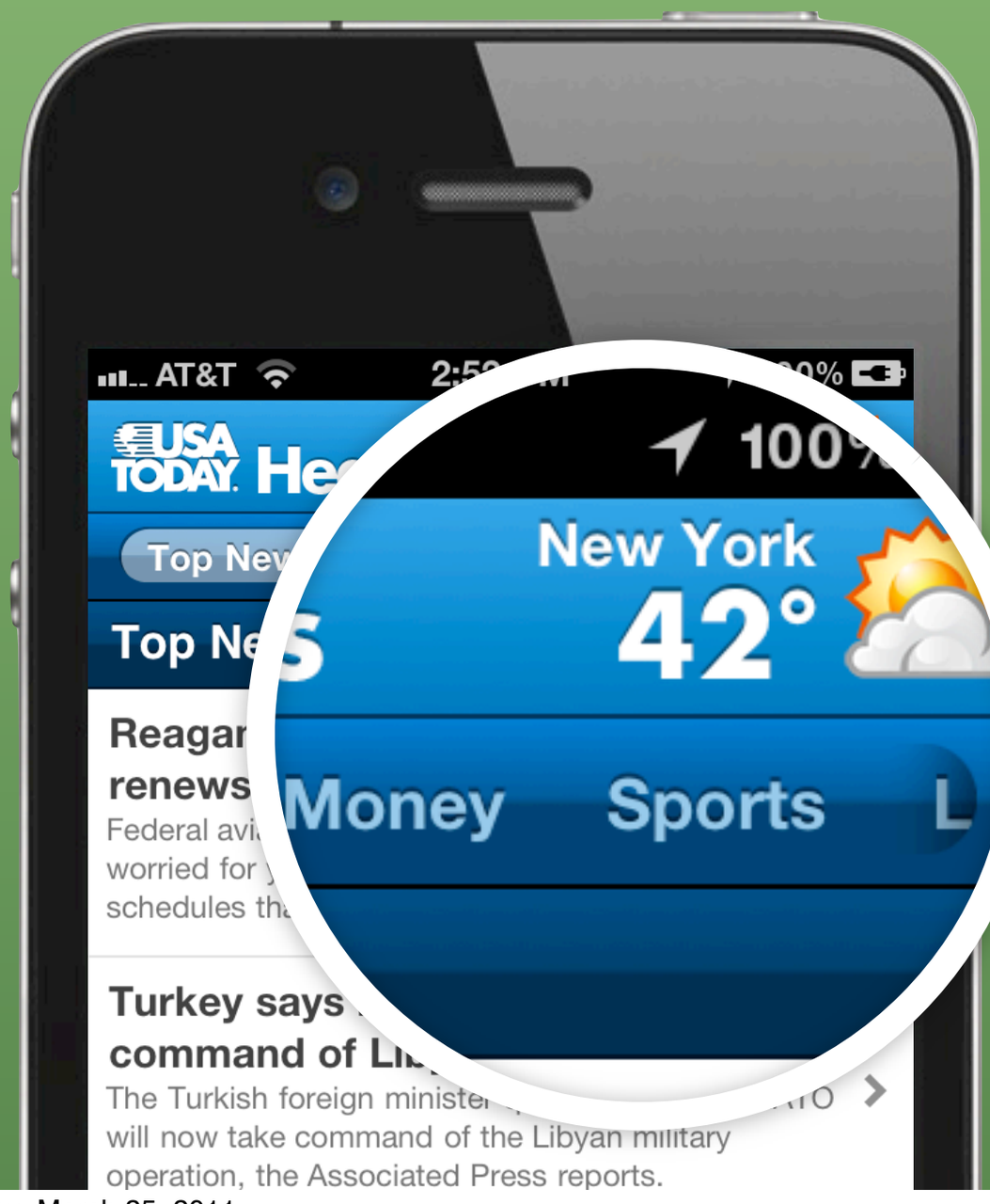


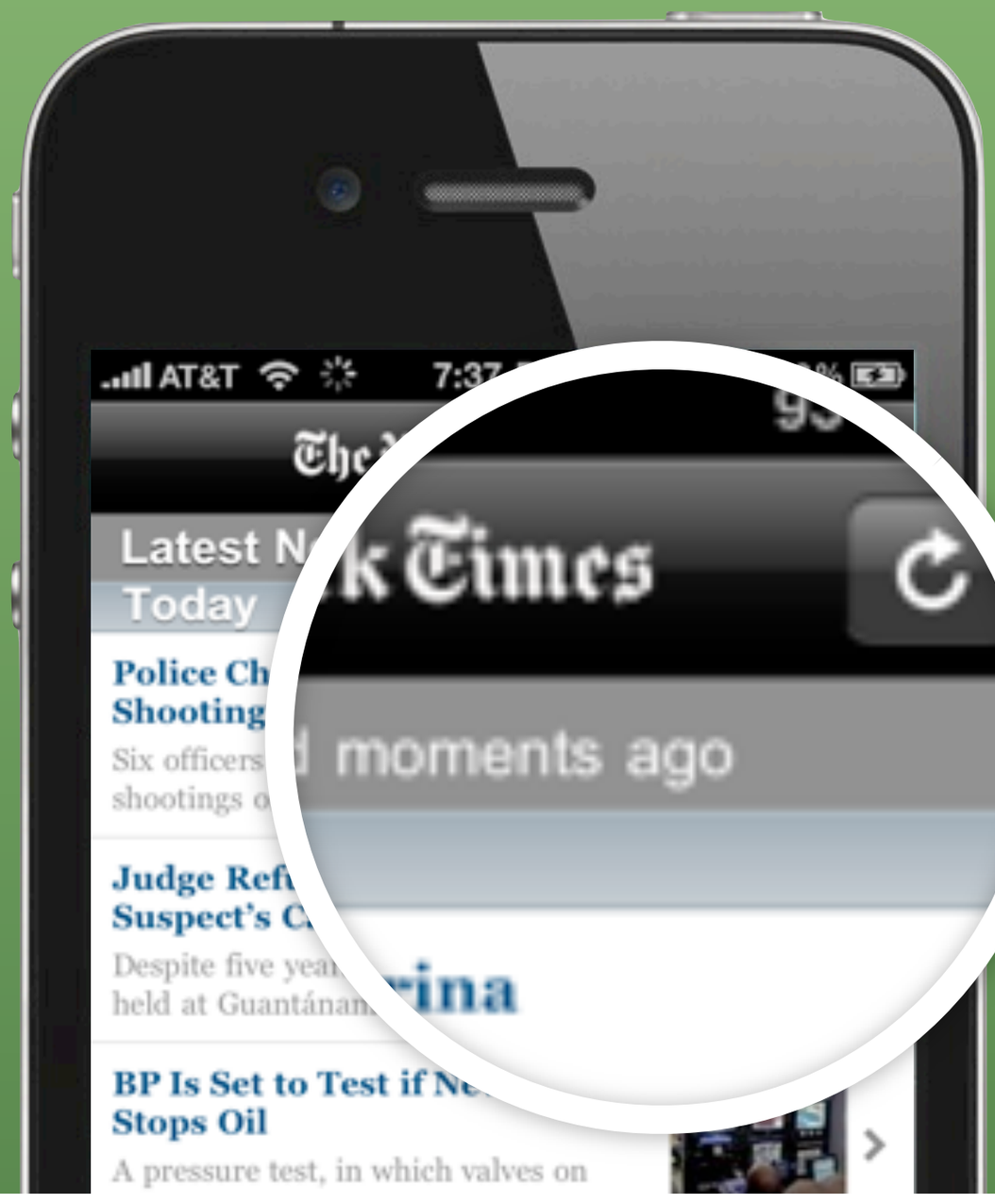
Cards vs Scroll



Affordances

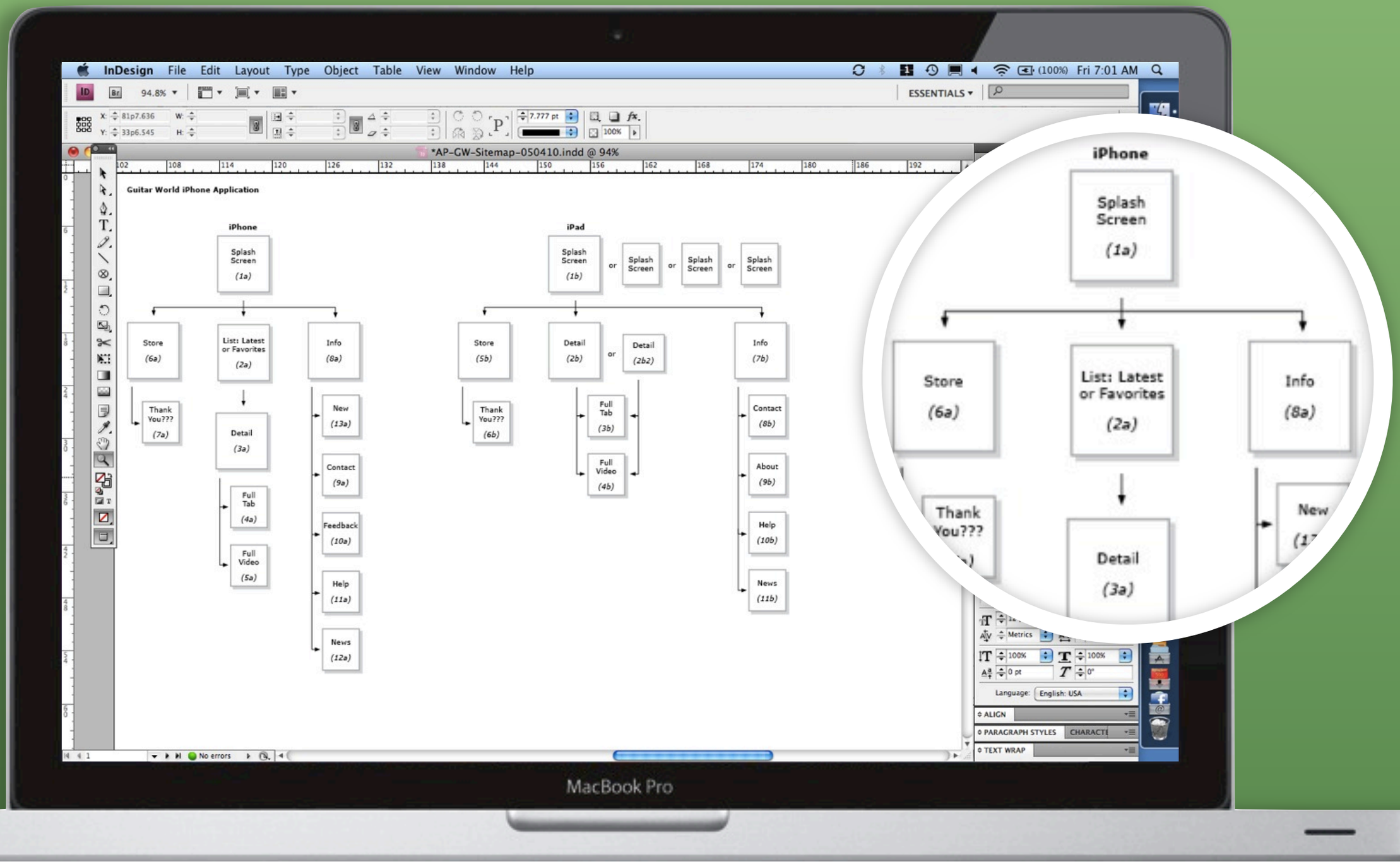
Be clear about what is clickable



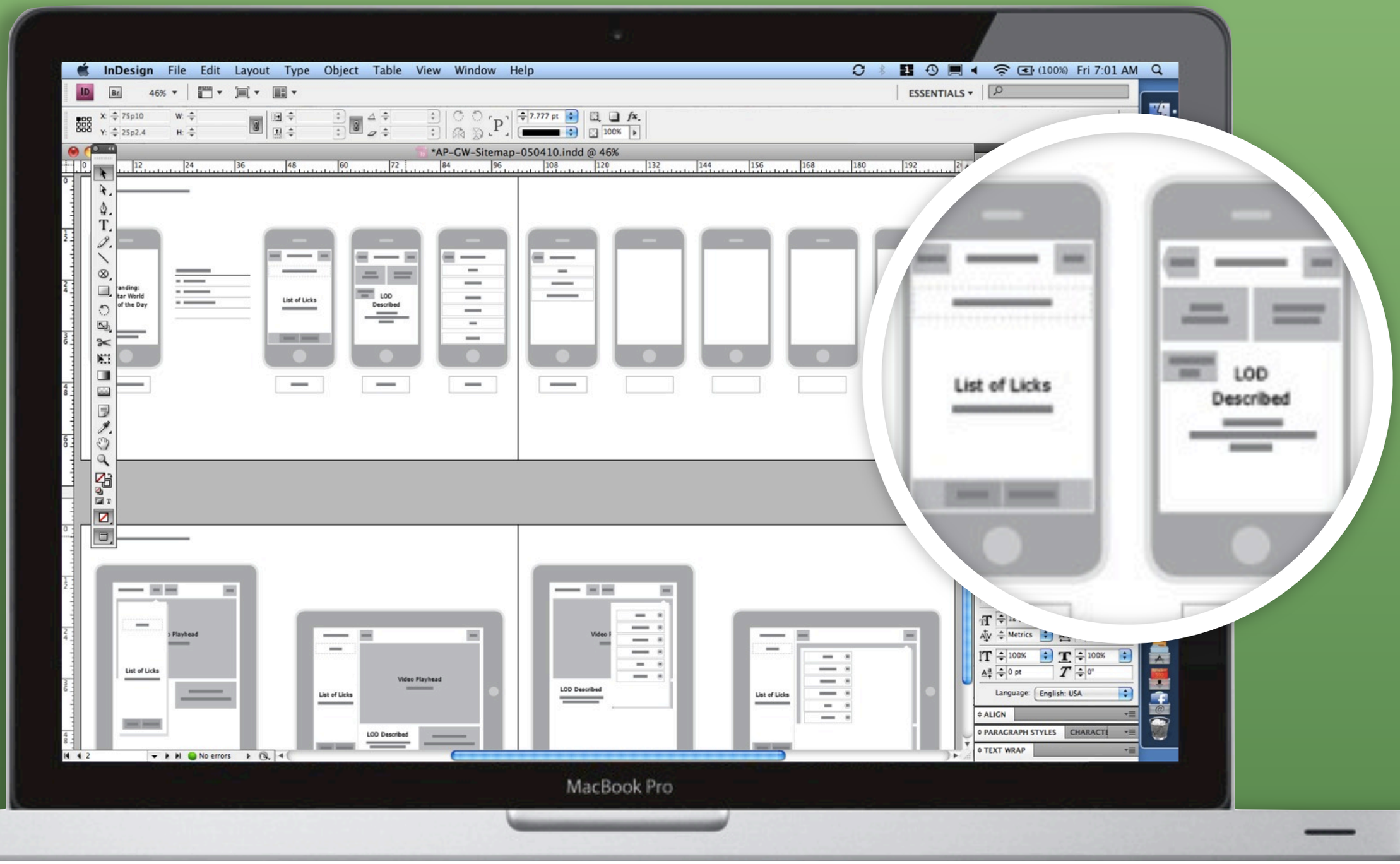


Information Architecture & Prototyping

Sitemaps


InDesign File Edit Layout Type Object Table View Window Help
*AP-GW-Sitemap-050410.indd @ 94%
Guitar World iPhone Application
iPhone
Splash Screen (1a)
Store (6a)
Thank You??? (7a)
List: Latest or Favorites (2a)
Detail (3a)
Full Tab (4a)
Full Video (5a)
Info (8a)
New (13a)
Contact (9a)
Feedback (10a)
Help (11a)
News (12a)
iPad
Splash Screen (1b)
or
Splash Screen
Store (5b)
Thank You??? (6b)
Detail (2b)
Detail (2b2)
Full Tab (3b)
Full Video (4b)
Info (7b)
Contact (8b)
About (9b)
Help (10b)
News (11b)
iPhone
Splash Screen (1a)
Store (6a)
List: Latest or Favorites (2a)
Info (8a)
Thank You???
Detail (3a)
New
MacBook Pro

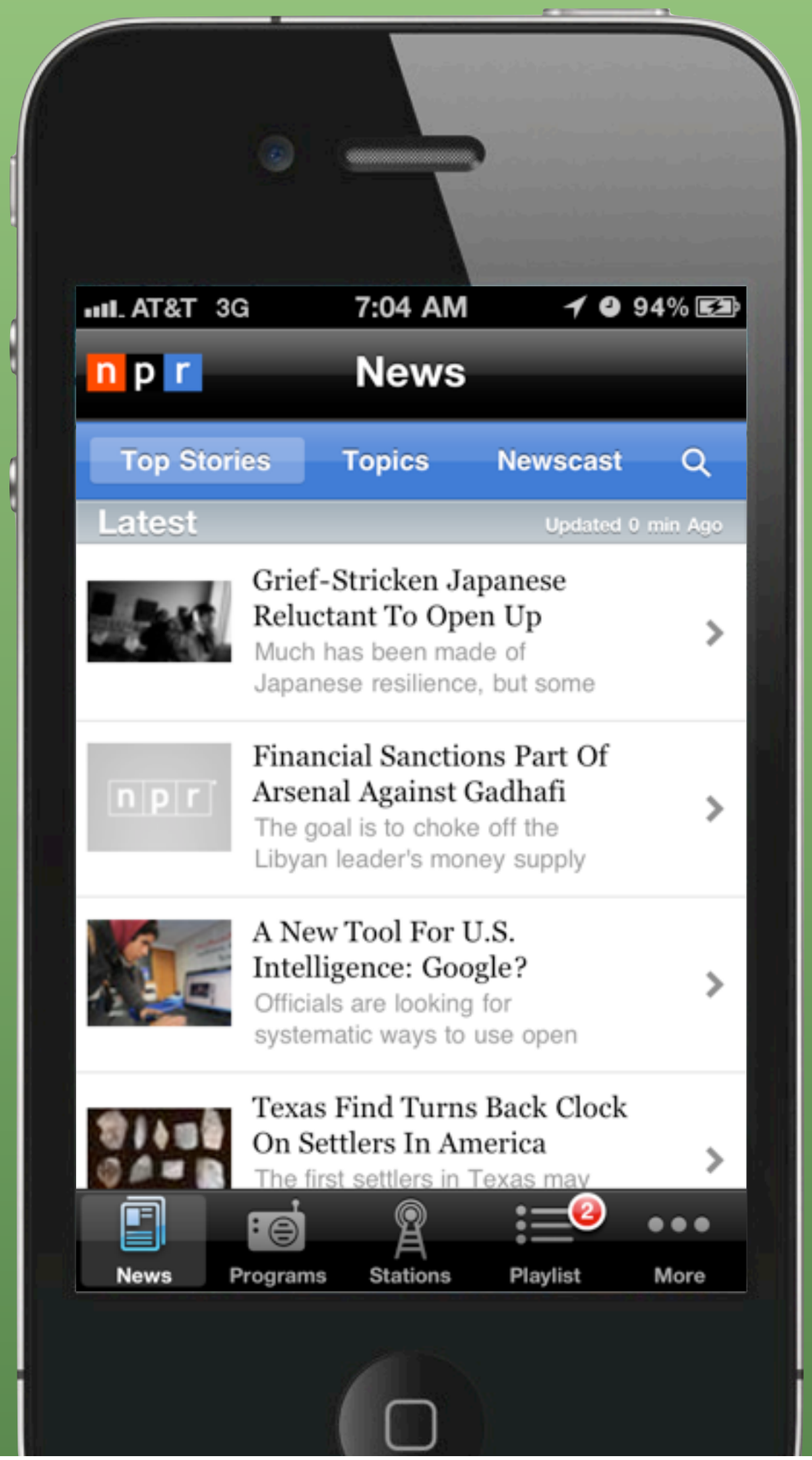
Wireframing



Wireframing

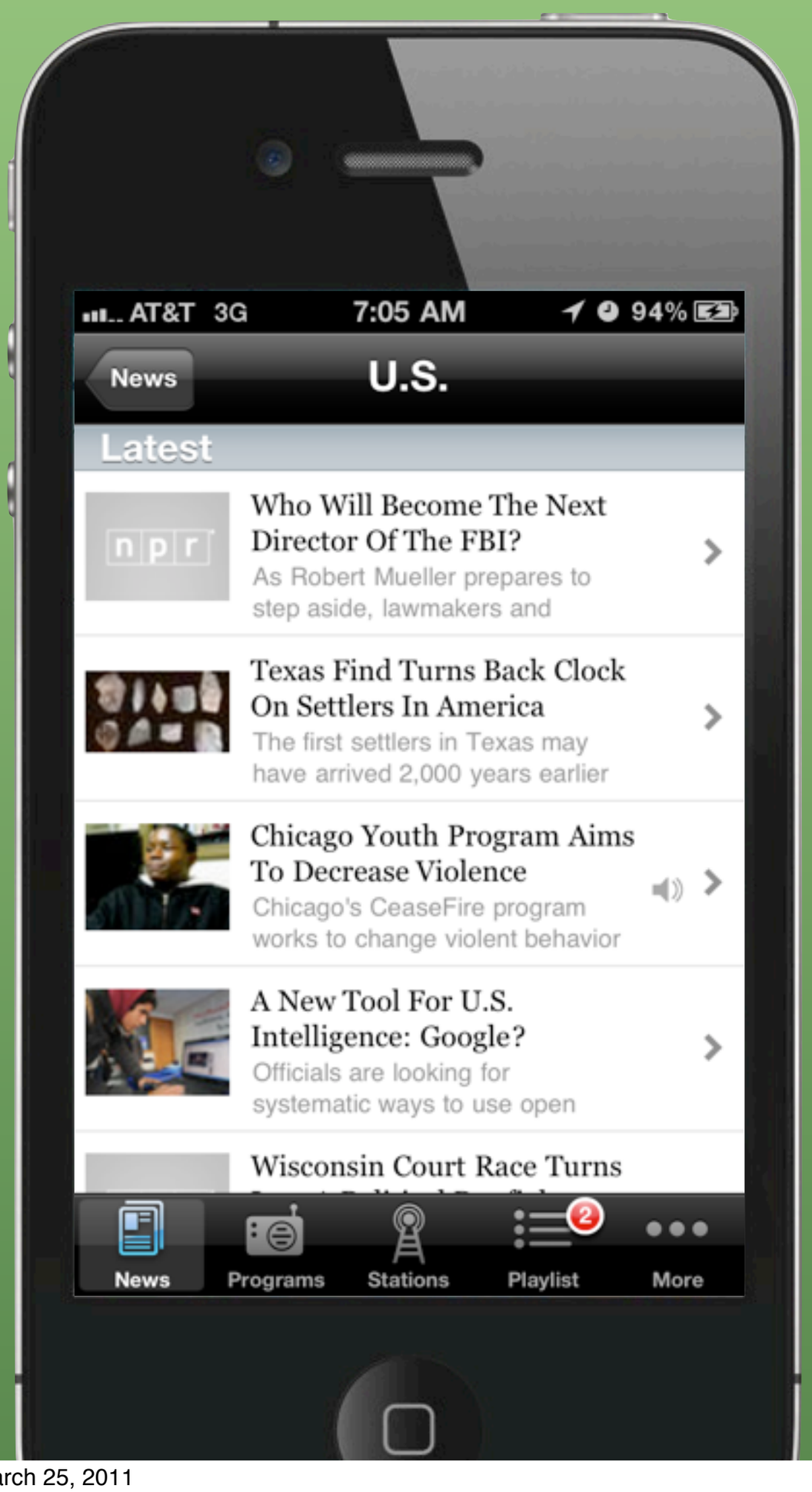
App	Link: bit.ly/SND-MobileDesign-Wireframe
Photoshop	bit.ly/SND-MobileDesign-Wireframe-Photoshop
Illustrator	bit.ly/SND-MobileDesign-Wireframe-Illustrator
Indesign	yhoo.it/SND-MobileDesign-Wireframe-Indesign
Stencil	bit.ly/SND-MobileDesign-Wireframe-Stencil
Web-Based	bit.ly/SND-MobileDesign-Wireframe-Browser
Keynote	bit.ly/SND-MobileDesign-Wireframe-Keynote
OmniGraffle	yhoo.it/SND-MobileDesign-Wireframe-OmniGraffle

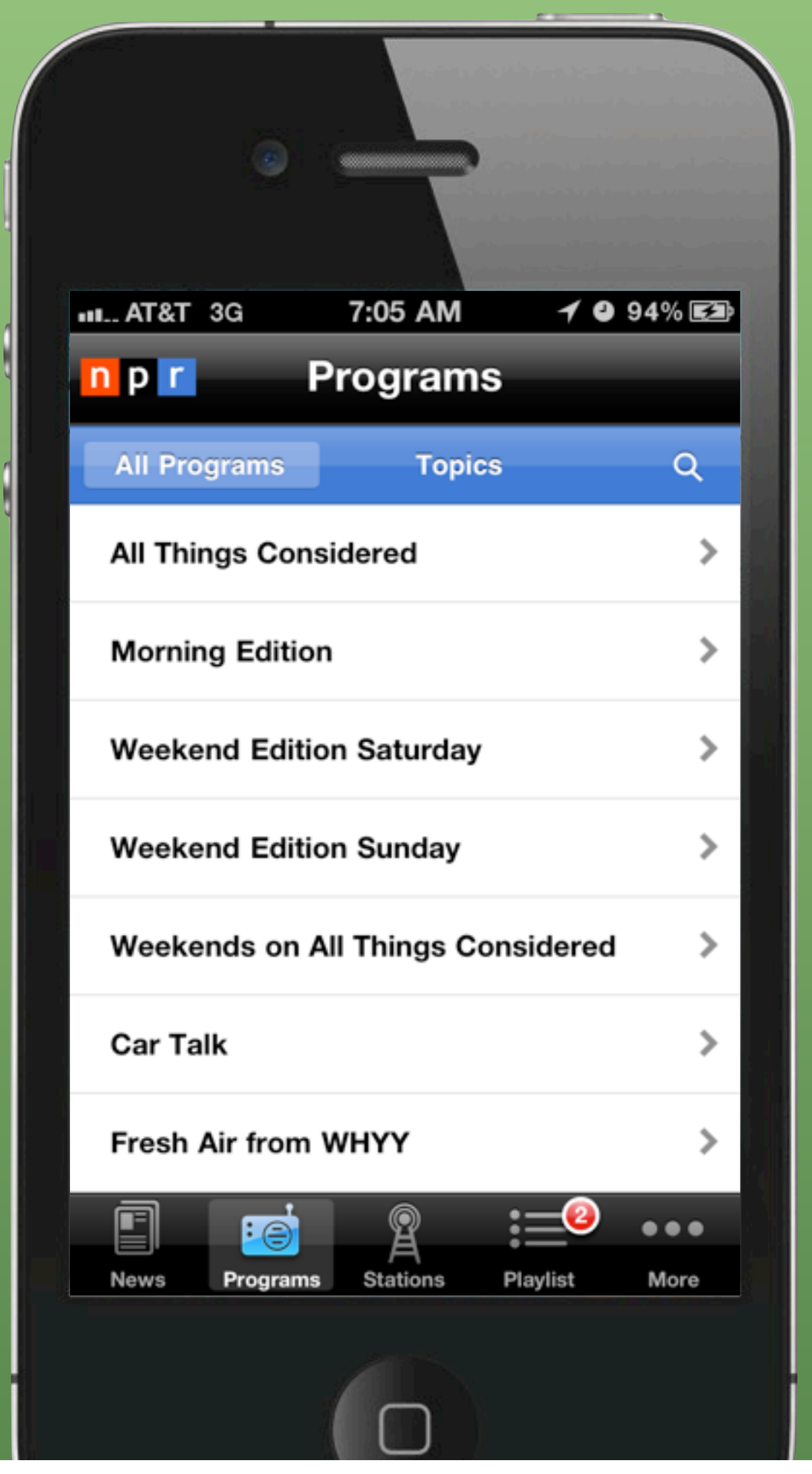

AT&T 3G
7:04 AM
94%
npr news
DEVELOPED BY bottle rocket

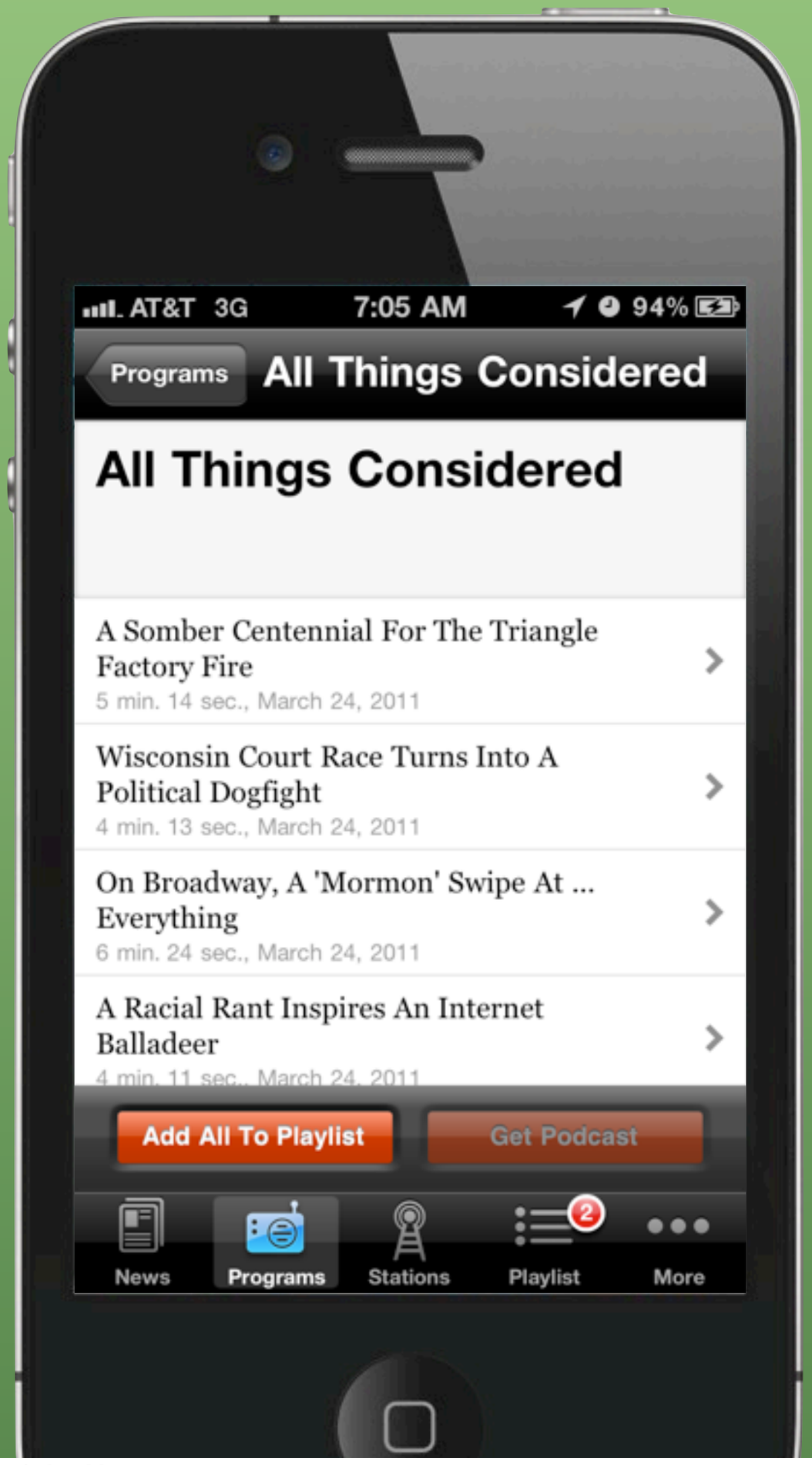

AT&T 3G
7:04 AM
94%
npr
News
Top Stories
Topics
Newscast
Latest
Updated 0 min Ago
Grief-Stricken Japanese Reluctant To Open Up
Much has been made of Japanese resilience, but some
Financial Sanctions Part Of Arsenal Against Gadhafi
The goal is to choke off the Libyan leader's money supply
A New Tool For U.S. Intelligence: Google?
Officials are looking for systematic ways to use open
Texas Find Turns Back Clock On Settlers In America
The first settlers in Texas may
News
Programs
Stations
Playlist
2
More


AT&T 3G
7:05 AM
94%
npr
News
Top Stories
Topics
Newscast
npr
hourly newscast
Hourly Newscast
by NPR
Mar 25, 2011
06:14 am CT
4 min. 46 sec.
Listen Now
A brief news update at the top of every hour.
News
Programs
Stations
Playlist
2
More

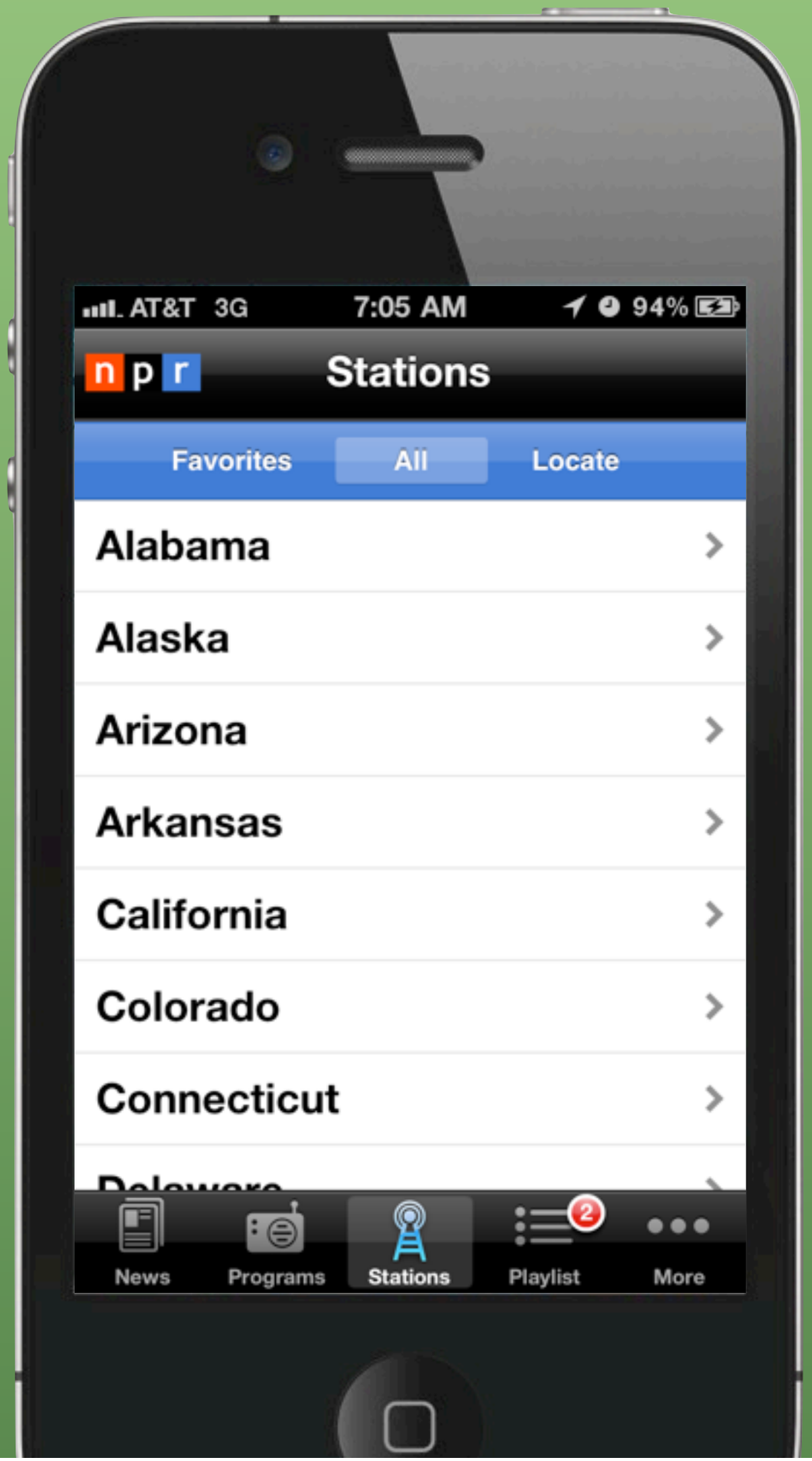

AT&T 3G
7:05 AM
94%
npr
News
Top Stories
Topics
Newscast
U.S.
World
Politics
Business
Music
Science
Health
News
Programs
Stations
Playlist
2
More

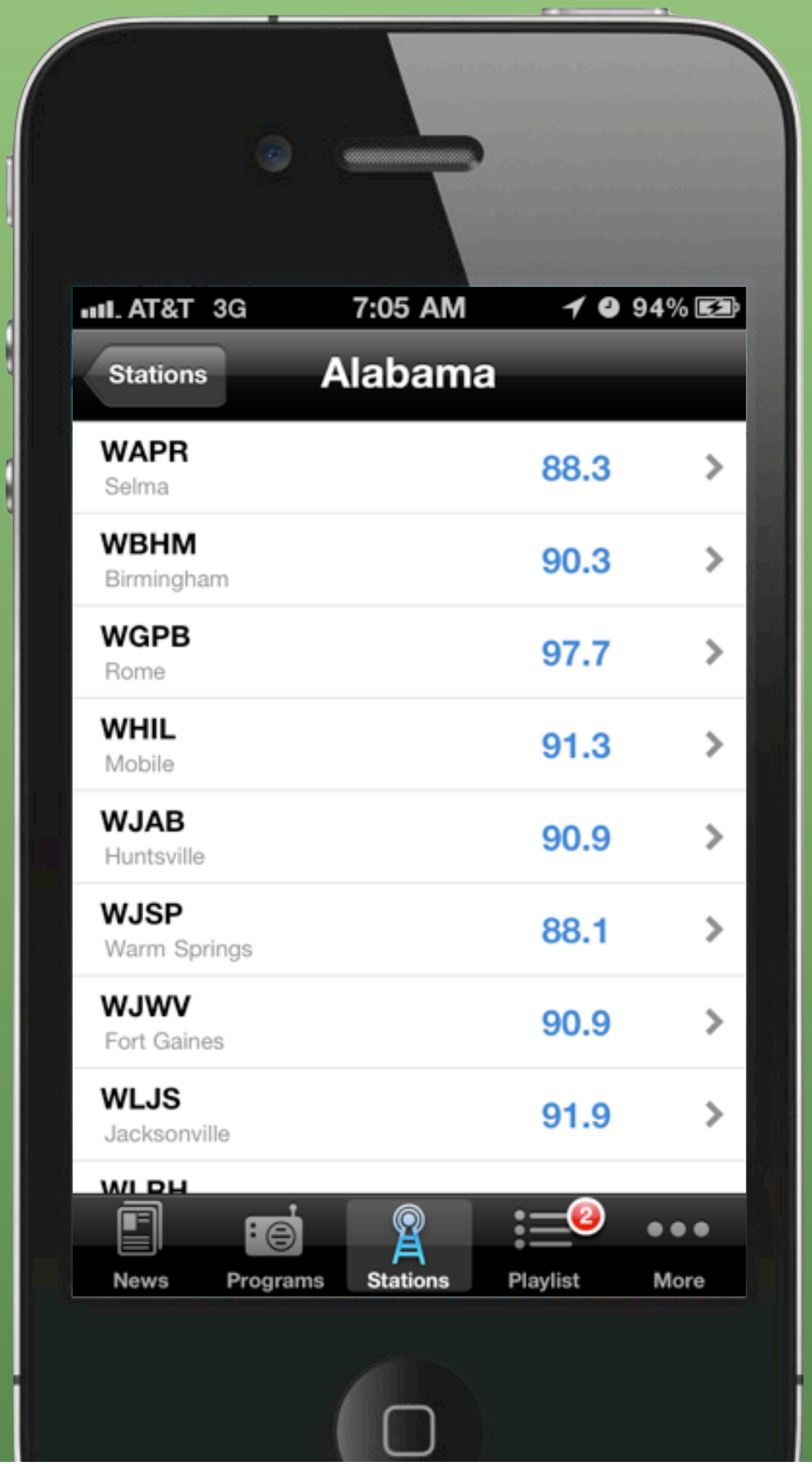

AT&T 3G
7:05 AM
94%
News
U.S.
Latest
Who Will Become The Next Director Of The FBI?
As Robert Mueller prepares to step aside, lawmakers and
Texas Find Turns Back Clock On Settlers In America
The first settlers in Texas may have arrived 2,000 years earlier
Chicago Youth Program Aims To Decrease Violence
Chicago's CeaseFire program works to change violent behavior
A New Tool For U.S. Intelligence: Google?
Officials are looking for systematic ways to use open
Wisconsin Court Race Turns
News
Programs
Stations
Playlist
2
More

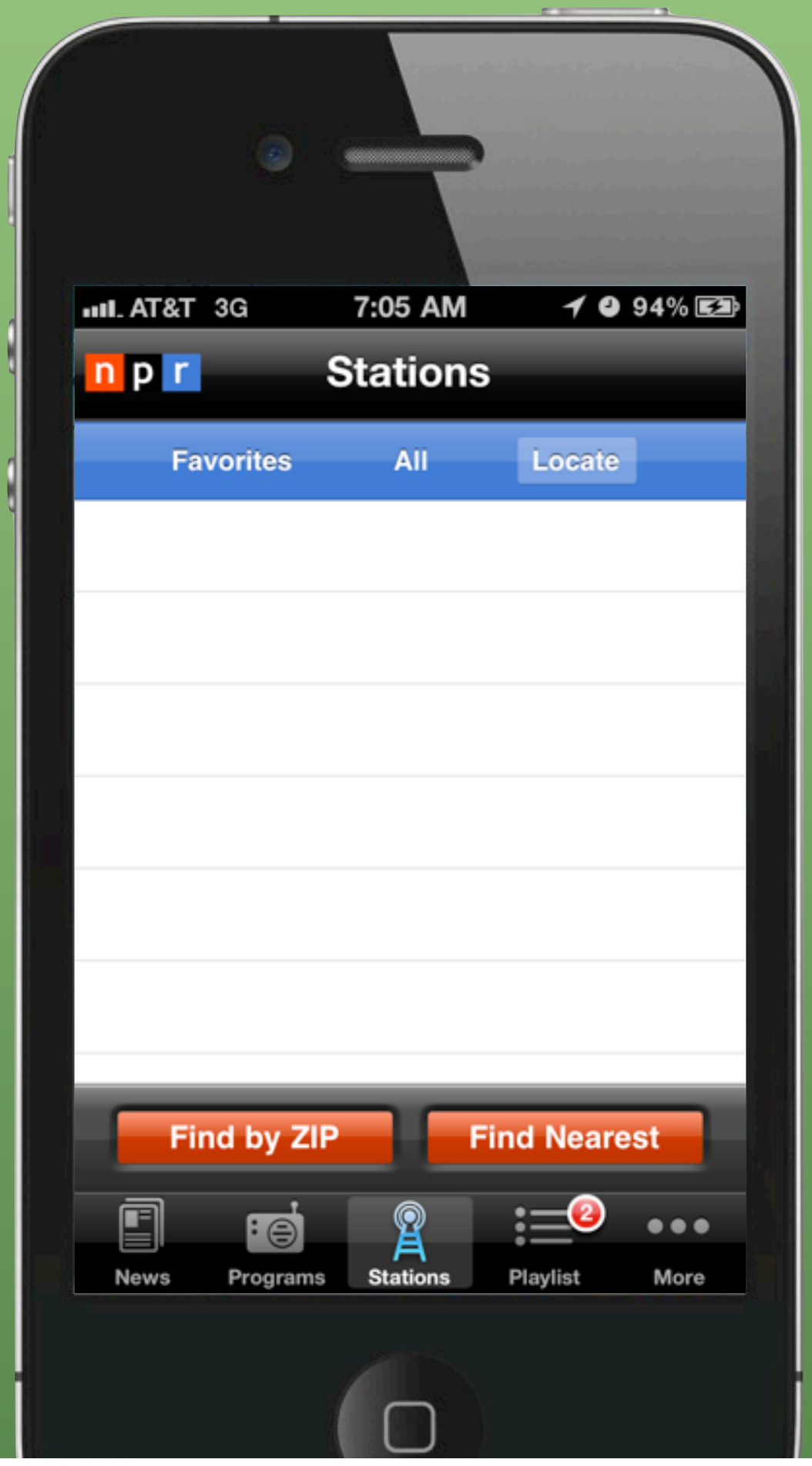

AT&T 3G
7:05 AM
94%
npr
Programs
All Programs
Topics
All Things Considered
Morning Edition
Weekend Edition Saturday
Weekend Edition Sunday
Weekends on All Things Considered
Car Talk
Fresh Air from WHYY
News
Programs
Stations
Playlist
2
More

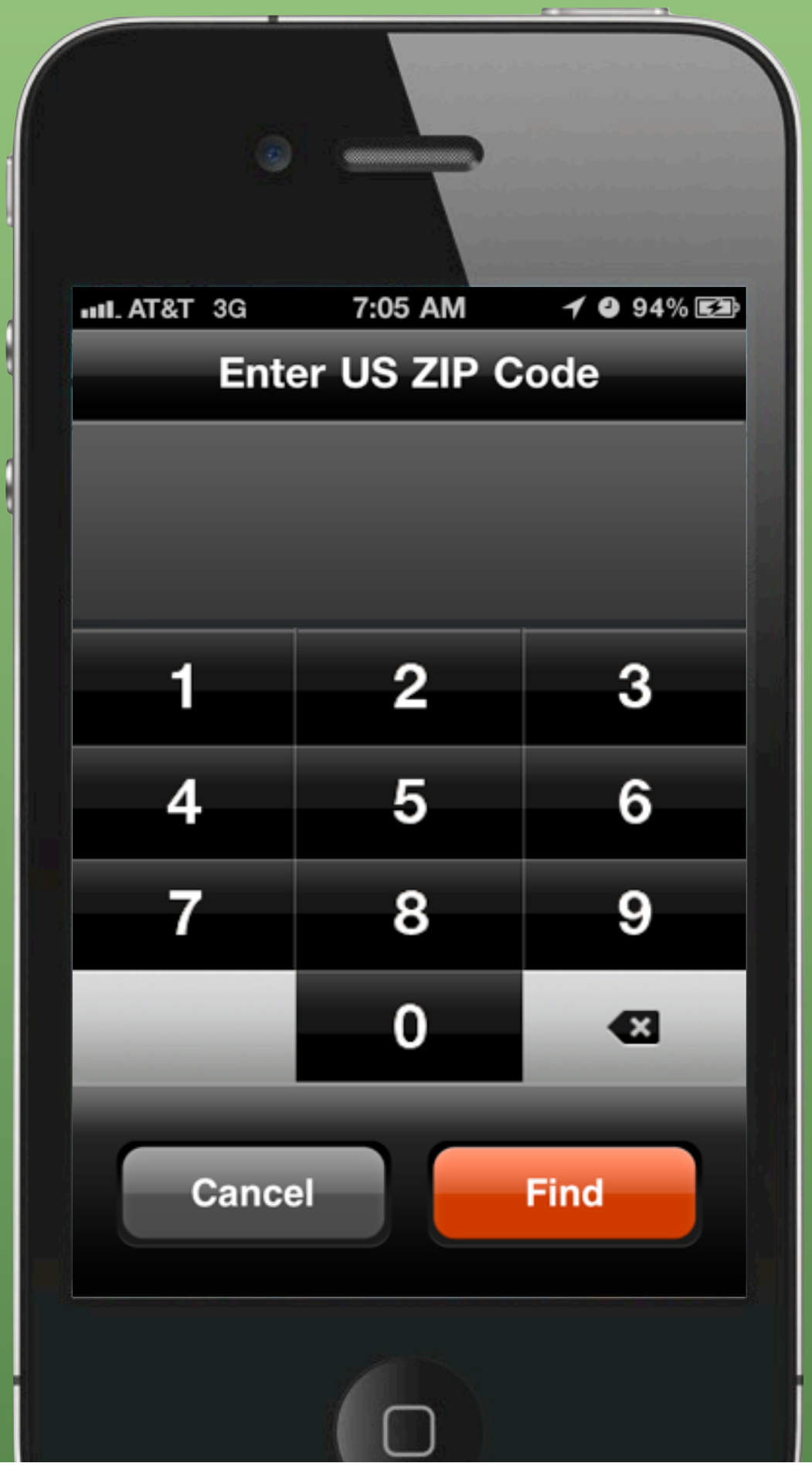

AT&T 3G
7:05 AM
94%
Programs
All Things Considered
All Things Considered
A Somber Centennial For The Triangle Factory Fire
5 min. 14 sec., March 24, 2011
Wisconsin Court Race Turns Into A Political Dogfight
4 min. 13 sec., March 24, 2011
On Broadway, A 'Mormon' Swipe At ... Everything
6 min. 24 sec., March 24, 2011
A Racial Rant Inspires An Internet Balladeer
4 min. 11 sec., March 24, 2011
Add All To Playlist
Get Podcast
News
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Stations
Playlist
2
More

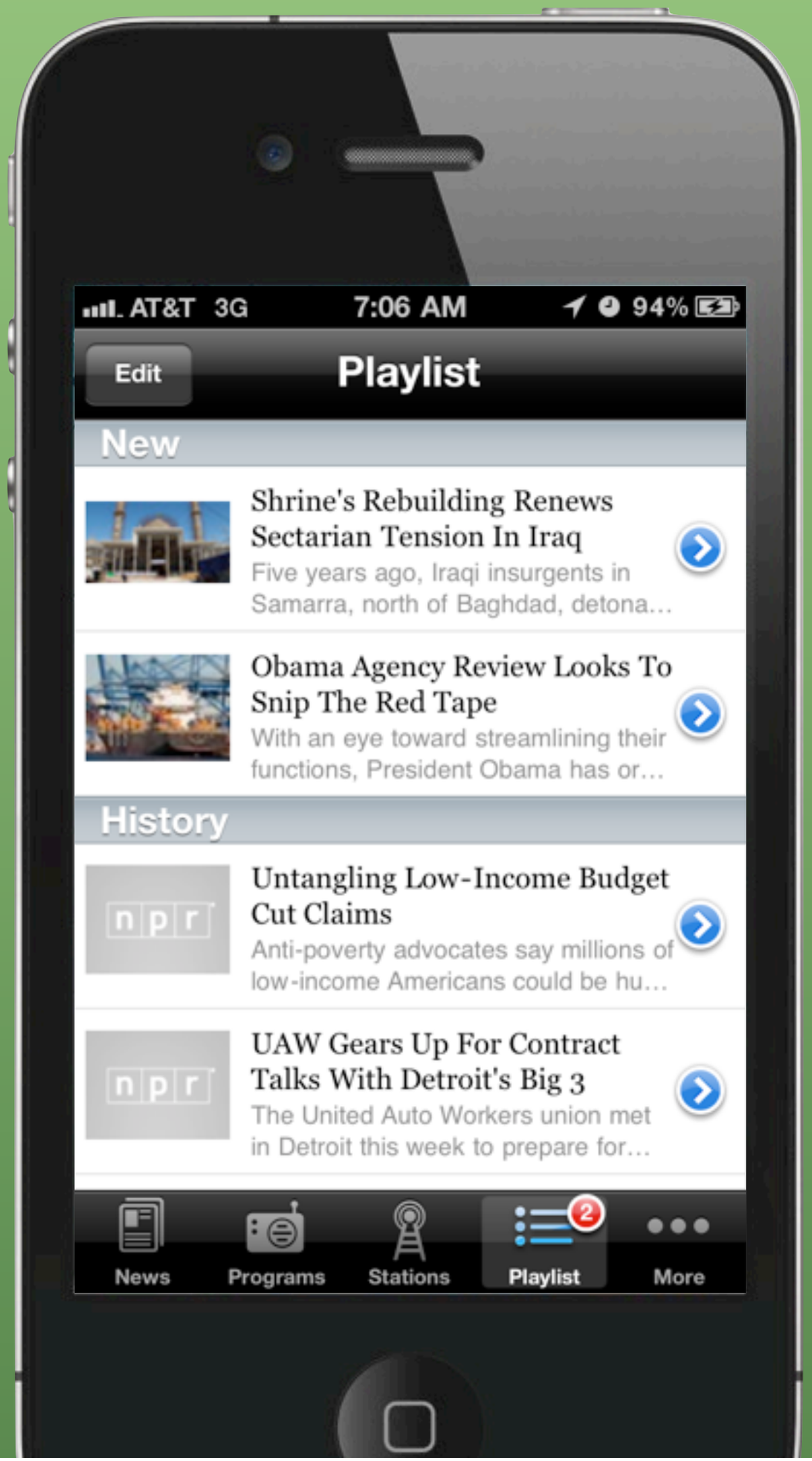

AT&T 3G
7:06 AM
95%
U.S.
By Topic
Who Will Become The Next Director Of The FBI?
by Carrie Johnson - March 25, 2011
Morning Edition
Share
Over the next several weeks, the Obama administration will have a big decision on its hands: choosing the next director of the FBI. The 10-year term of the FBI's current leader, Robert Mueller, expires in September. A committee is already searching for candidates to replace him.
Mueller became the FBI director in September 2001, days before the terrorist attacks that would change the bureau's
News
Programs
Stations
Playlist
2
More

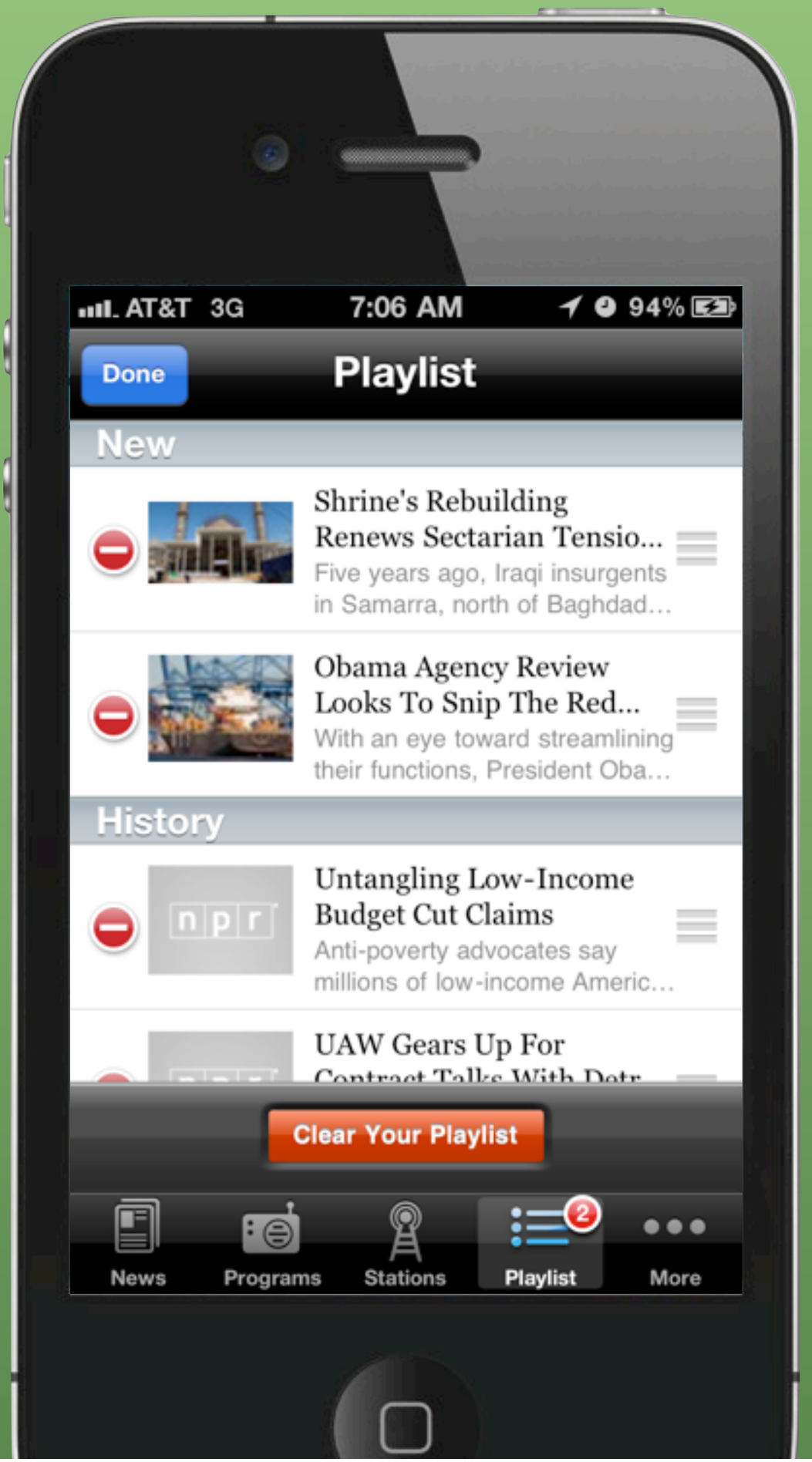

AT&T 3G
7:05 AM
94%
Stations
Favorites
All
Locate
Alabama
Alaska
Arizona
Arkansas
California
Colorado
Connecticut
News
Programs
Stations
Playlist
2
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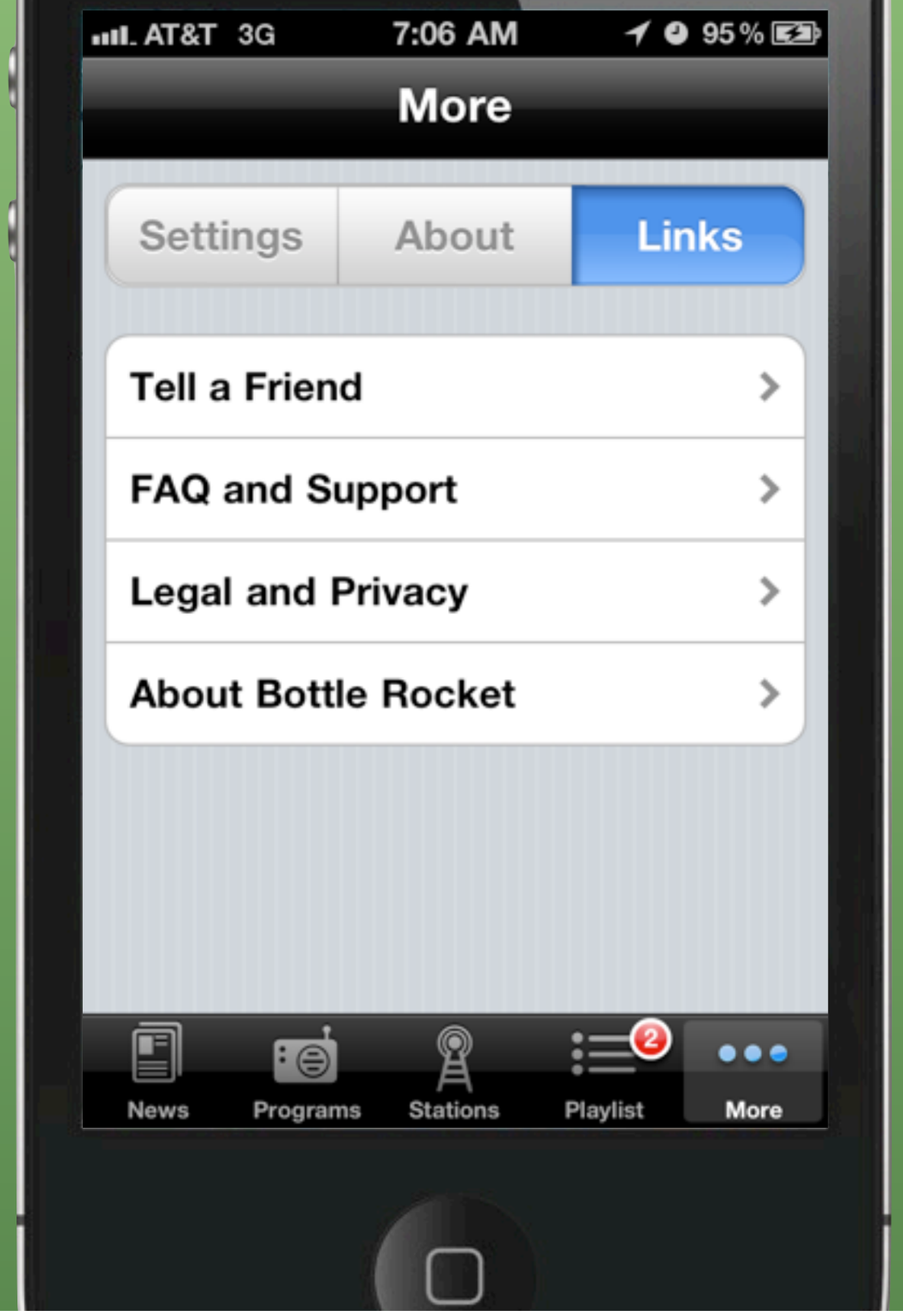

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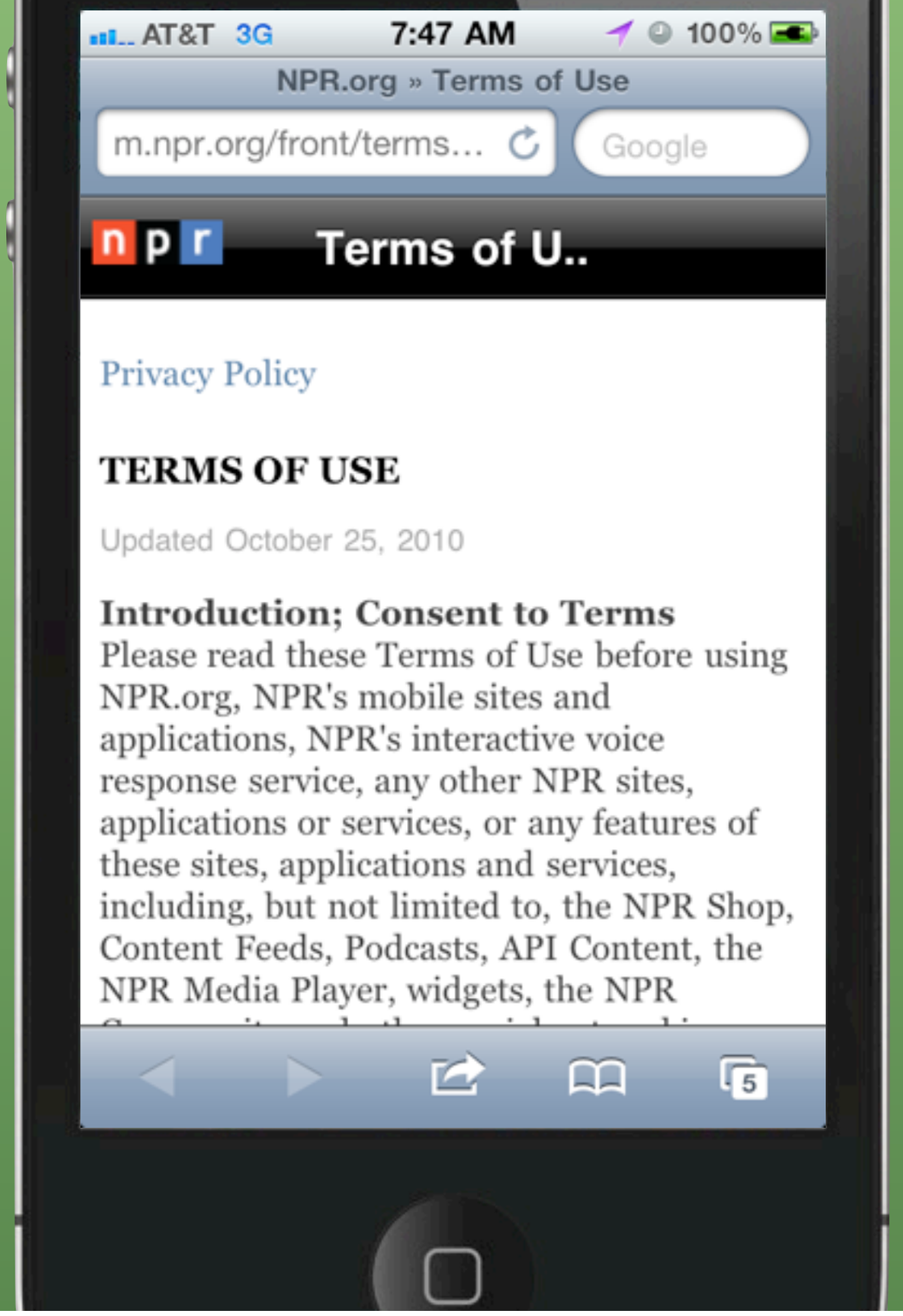

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